



A review on semi-supervised clustering

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ABSTRACT

Semi-supervised clustering (SSC), a technique integrating semi-supervised learning and clustering analysis, incorporates the given prior information (e.g., class labels and pairwise constraints) into clustering to guide the clustering process and improve the performance. In recent years, a large number of valuable works have emerged, focusing on theoretical research and application in different fields. In this paper, a detailed review of SSC is provided from a new perspective. Firstly, all SSC studies are organized as partition-based SSC, hierarchical-based SSC, density-based SSC, graph-based SSC, neural network-based SSC, Nonnegative Matrix Factorization-based SSC and random subspace technique-based SSC. Thus, the semi-supervised researches can be in-depth discussed in each clustering idea. Secondly, the general overviews are detailed in each category respectively, including the performance, the suitable scenarios and the way to add supervising information. Thirdly, the recent successful applications of SSC are summarized according to different backgrounds such as medical, biological, business, journalism, financial and so on. Based on this, some application caveats and development trends of SSC are particularly given in the end. This comprehensive review and analysis of SSC can provide an overall outline, the scope of research topics, and a relative complete analysis of existing SSC methods for researchers.

1. Introduction

Semi-supervised clustering (SSC) is a new research direction in the field of machine learning and an essential branch of data mining in recent years. This novel semi-supervised clustering learning technique, which combines semi-supervised learning with cluster analysis, has been widely used in many practical fields [25], such as bioinformatics, text clustering, etc.

1.1. Clustering, semi-supervised learning and semi-supervised clustering

Clustering, an efficient technique to mine useful information, divides a set of objects into clusters according to their similarity [172], [150]. Conventional clustering only utilizes unlabeled data for training a model. This model conversely predicts these unlabeled data. It generally includes partition-based methods, hierarchical-based methods, density-based methods and graph-based methods [146]. These methods often encounter the problem of high inconsistency between clustering result and actual partition as the ignorance of prior knowledge.

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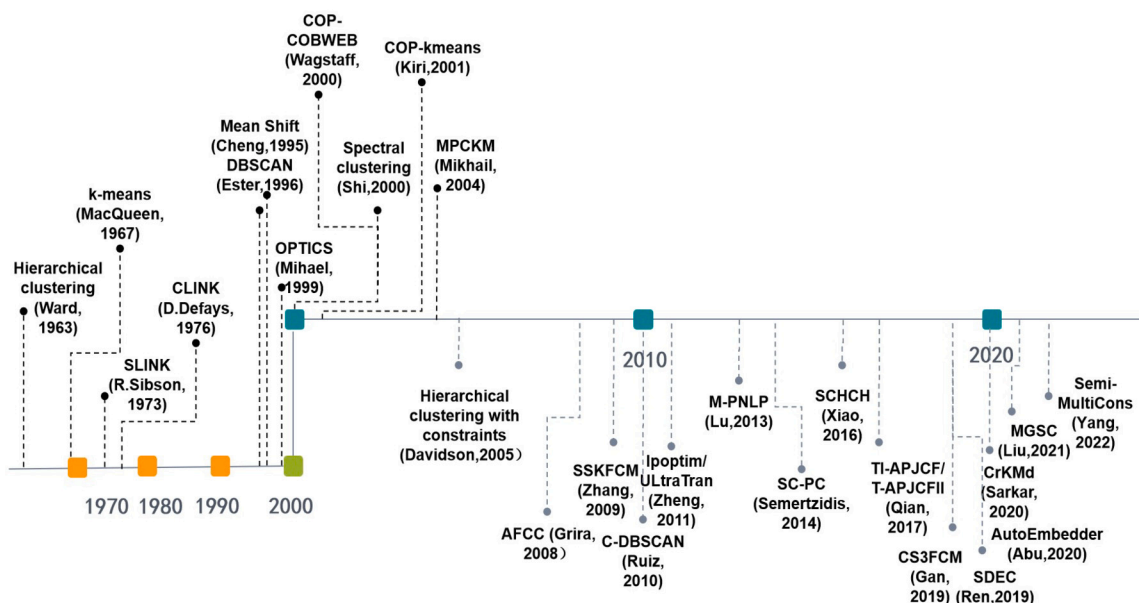


Fig. 1. Development timeline of representative unsupervised clustering and some semi-supervised clustering. (The methods above the line have a higher citation rate, and the methods with a lower citation rate are below. The length of the line does not mean anything.)

Since conventional clustering does not use supervising information, it is an unsupervised learning technique. There are three learning techniques (unsupervised learning, supervised learning and semi-supervised learning (*SSL*)) in the field of machine learning [28], [147], which are judged by whether they have supervising information. In some cases, tagged data accounts for a small proportion, the relatively limited tagged data is not enough to provide enough data distribution information for supervised learning to ensure the generalization ability of the method [54], [112]. However, unlabeled data is abundant and can be easily obtained. To tackle the imbalance among data volume and the quantity of tags, semi-supervised learning emerged. *SSL* uses model assumptions on the data distribution to build a learner to pass the existing label the unlabeled samples [170].

Conventional unsupervised clustering ignores certain domain background knowledge that users have (assume that the cluster labels of all sample data are unknown). Therefore, researchers aim to apply this domain knowledge to the clustering task, which is called semi-supervised clustering. Different from conventional unsupervised clustering, *SSC* aims to utilize a limited number of labeled data to predict the cluster tag of the unlabeled data by learning a cluster model using both unlabeled data and labeled data. Apart from that, the obtained semi-supervised clustering model can predict test data. Consequently, semi-supervised clustering gradually attracts more attention and becomes a new research hotspot.

In the 1960s, the emergence of hierarchical clustering promoted the development of unsupervised clustering. In the next few years, many unsupervised clustering methods were proposed, such as *k-means*, [92], *SLINK*, [6], *CLINK*, [69], *Mean Shift*, [27], *OPTICS*, [3] (full names corresponding to the abbreviation of the specific methods in this paper are provided in the Appendix). Semi-supervised clustering was developed around 2000. The development timeline of representative unsupervised clustering and part of the SSC methods is displayed in Fig. 1. Early researches on unsupervised clustering set the stage for most subsequent research in SSC. In the early stages of the development in SSC, prior information was employed to instruct clustering directly. While the prior information may be negative, some researches [166], [39] considered the safe confidence to obtain great clustering. In 2000, spectral clustering obtained great success in graph-based clustering. Since then, spectral clustering is widely applied in semi-supervised graph clustering in both single graph and multiple graphs. With the combination of deep neural networks and clustering, clustering performance is further improved. *AutoEmbedder*, [104] and *SDEC*, [113] have been further improved on the traditional deep clustering. More recently, with the increasingly wide application, the data to be processed is more complex and the dimension is higher. Many Nonnegative Matrix Factorization (*NMF*)-based and random subspace technique-based semi-supervised clustering methods, such as *PGCNMF*, [143], *DNMF*, [4], *DCECP*, [162], *MVOCNMF*, [50], were proposed to adapt the demand of high-dimensional data.

1.2. Motivations and contributions

With the proposal of a considerable amount of SSC methods, some researchers embark on SSC review. In the early reviews [43], [30], [128], [120], some representative methods were described. However, due to the rapid development of semi-supervised clustering technology, recent important algorithms are not available in these surveys. Besides, some reviews discuss the relatively new methods, while these reviews are not making a comprehensive overview. Such as [38] only reviews semi-supervised fuzzy kernel clustering, [105] summarizes existing works which are designed for handling high-dimensional sparse data, [28], [20] and [78] respectively focus on some research progress of graph-based, density-based or safe semi-supervised learning, while graph-based, density-based and safe semi-supervised clustering methods in their review only occupy a small portion. [110] attempts to analyze

semi-supervised clustering comprehensively, while the taxonomy is quite different from ours. It is directly organized by the way to use supervising information, which lacks the effect of applying different supervised processes on various clustering methods. In addition to this, the existing reviews do not analyze the evaluation metrics of semi-supervised clustering. The above motivations motivate us to conduct a comprehensive review of existing semi-supervised clustering methods and to propose a new organization to discuss the performance of each class of methods as well as the direction of development. This comprehensive review and analysis of SSC can provide researchers with an overall outline, scope of the research topic, and a relatively complete analysis of existing SSC methods.

By analyzing previous and recent studies on SSC, we provide a comprehensive survey on SSC. The SSC methods in this survey is organized from the perspective of different clustering methodologies [136] (partition-based SSC, hierarchical-based SSC, density-based SSC, graph-based SSC, neural network-based SSC, Nonnegative Matrix Factorization-based SSC and random subspace technique-based SSC). For the methods in each subsection, the fundamental theories are reviewed, the performance and the suitable scenarios are compared, and the way to use supervising information are also dissected. Compared with the review analyzed directly according to the use of supervising information, our organization is more fine-grained, and it can reflect the distinction between various ways of using supervising information under different clustering methods. The applications of SSC methods in different fields are also expounded and the challenges in each field are summarized. Besides, we discuss several caveats and development trends for SSC. Through the comprehensive review and the analysis of SSC, this paper aims to provide an overall outline and a complete analysis of existing SSC methods for beginners and the scope of research topics for researchers. It can also promote the development of more effective SSC technologies as well as the significant researches in this domain. In summary, our contributions are listed as follows.

- 1) **A comprehensive review of SSC models.** We organize existing SSC methods referring to the organization method of conventional clustering, and summarize existing SSC methods according to prior information and the way to add prior information. The proposed taxonomy helps beginners to quickly grasp the frontier ideas of SSC and researchers to gain a deeper understanding of the crucial characteristics of SSC models.
- 2) **A performance evaluation and analysis of SSC models.** We provide a detailed comparative analysis of SSC methods in terms of time complexity, evaluation index, performance, and further research the evaluation metrics for evaluating the performance of the SSC models. By evaluating existing SSC methods, we reveal the strengths and weaknesses and applicable scenarios of semi-supervised clustering methods, which open a promising direction for forthcoming efforts.
- 3) **An overview of open questions and future directions.** We analyze the problems solved by semi-supervised clustering in different fields and reveal potential directions in each field for future research. We also thoroughly review several fundamental issues regarding semi-supervised clustering model design, supervising information collection, etc., and elucidate potential directions for future research.

The rest of this survey is organized as below. (1) The organizational structure analysis of existing SSC reviews is given in Section 2. (2) The research progress of SSC methods is reviewed in Section 3. (3) The evaluation measures for SSC are analyzed in Section 4. (4) The applications of SSC are described in Section 5. (5) The challenges and development directions for SSC are given in Section 6. (6) The conclusions are given in Section 7.

2. Organizational structure analysis of semi-supervised clustering reviews

The semi-supervised clustering (SSC) can be organized based on the following three aspects (as Fig. 2 shows): (1) the conventional clustering methods, (2) the prior information, (3) the approaches to the use of supervising information.

- (1) Referring to the conventional clustering, the SSC can be organized into the following several types: partition-based SSC, hierarchical-based SSC, density-based SSC, graph-based SSC, neural network-based SSC, Nonnegative Matrix Factorization-based SSC and random subspace technique-based SSC.
 - *Partition-based SSC*: partition-based SSC, derived from traditional partition-based clustering, organizes n data objects into k ($k \leq n$) partitions (or clusters) by optimizing the objective function under the guidance of supervising information. The partition-based SSC includes the k -means model, FCM model and k -modes model.
 - *Hierarchical-based SSC*: hierarchical-based SSC uses agglomerative or divisive approaches to form the data into a hierarchical tree, it also incorporates the prior information into the construction of the hierarchical tree. Then the clusters can be obtained by traveling the hierarchical tree.
 - *Density-based SSC*: in the density-based SSC, an effective density matrix is learned under the assistance of prior knowledge to estimate the tightness of the sample distribution.
 - *Graph-based SSC*: graph-based SSC utilizes the connected edges to learn the label information of unlabeled nodes. This section includes single graph and multiple graphs clustering.
 - *Neural network-based SSC*: it represents a category of methods that aims to guiding prior information into the framework of the neural networks.
 - *Nonnegative Matrix Factorization-based SSC*: *Nonnegative Matrix Factorization* is a matrix factorization method, the decomposed matrices are non-negative, and the nonlinear dimension reduction is realized at the same time. Nonnegative Matrix

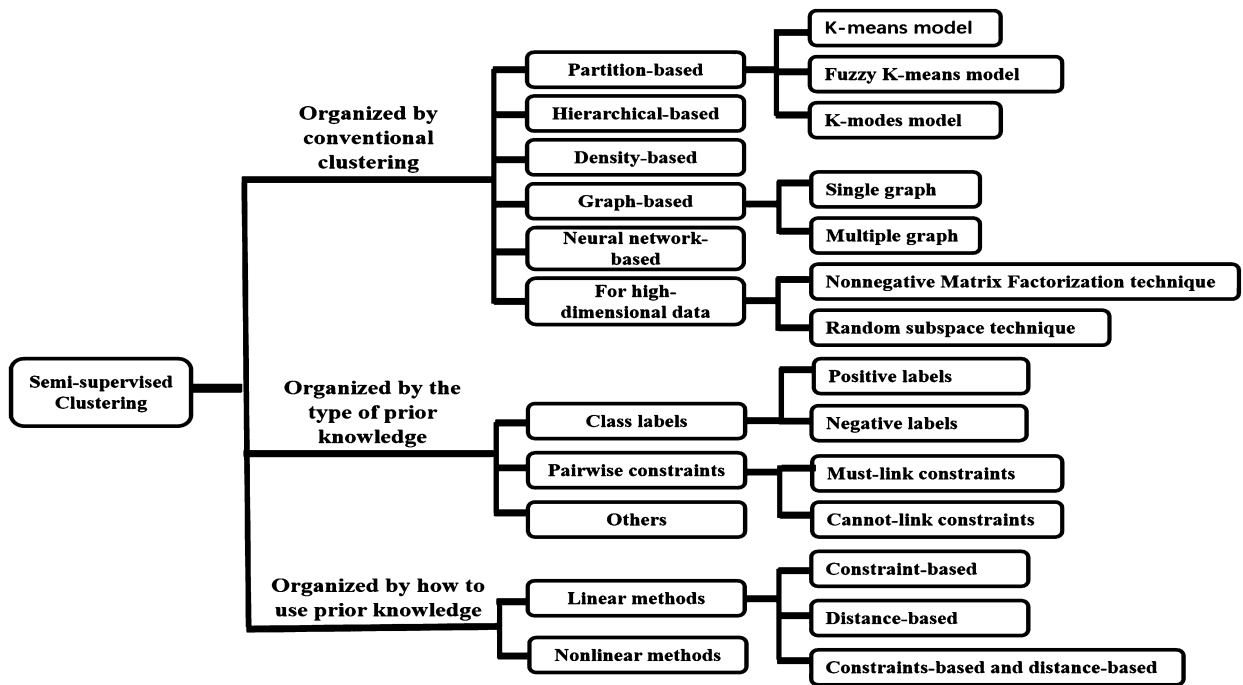


Fig. 2. Organizational structure analysis of semi-supervised clustering methods.

Factorization-based SSC applies the supervising information to the process of matrix factorization. These methods can solve the problem caused by high data dimension.

- *Random subspace technique-based SSC*: As the data dimension becomes higher and higher, these SSC methods perform random subspace technique before clustering.
- (2) SSC can be also organized according to the prior information [128]. The mainstream of the prior knowledge concludes class labels and pairwise constraints. Other types of prior information also appear, for instance the prior membership degree [158] and the grouping information [109].
- *Class labels*: class label represents which cluster the object belongs to, it contains positive and negative label.
 - *Pairwise constraints*: pairwise constraints contain the following two constraints (1) must-link constraints: it indicates that a pair of data has to be allocated to one cluster, (2) cannot-link constraints: it means the paired data cannot be allocated to one cluster. Since the efficiency of the methods based on pairwise constraint may easily affected by the size of prior information, researchers are dedicated to generating pairwise constraints to obtain better clustering as much as possible. Pairwise constraints are selected using active learning in [144], however, the existing methods for selecting pairwise constraints require labeled data. [96] solves this problem by using unlabeled data, it obtains a dense group of samples to select pairwise constraints after initializing the cluster.
 - *Others*: for the prior information of prior membership degree and the grouping information, the former is used for semi-supervised fuzzy c-means [109], while the latter is often used in the image segmentation [160].
- (3) How to use supervising information is another basis SSC, [105], [110]. It can be grouped into three kinds:
- *Constraint-based SSC*: these methods utilize supervising information to constrain the process of clustering and guide the method to obtain a better clustering result [9].
 - *Distance-based SSC*: distance-based methods often train a similarity measure to meet the label or constraint information, and cluster the data objects applying the trained similarity measure [8].
 - *Constraints-based and distance-based SSC*: their strategies are to use both the two methods simultaneously [14].

These three organizational bases organize SSC methods from different angles. By prior information and the way to use supervising information are mentioned in [109]. The prior information is generally given in terms of class labels or pairwise constraints. However, new forms of prior information, such as the cluster-level constraints [31] given in hierarchical-based SSC, have been tried to be proposed by researchers to meet different needs. Since the prior information is not fully available, it is not reasonable to classify SSC according to the type of prior information. Although by the way to use supervising information is feasible, it does not show the effect of different supervision approaches on various clustering methodology. To solve the problem mentioned above, this survey provides the first organizational basis, which is organized referring to the organization method of conventional clustering, to review the recent achievements on SSC.

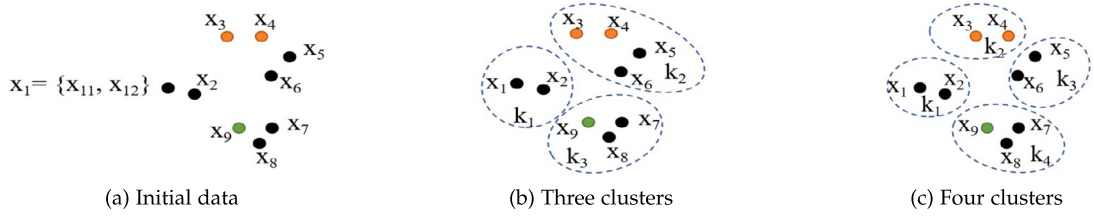


Fig. 3. An example of semi-supervised clustering process. (The black sample points indicate that the sample is not labeled. Colored sample points indicate that the sample is labeled, and different colors represent different label information.)

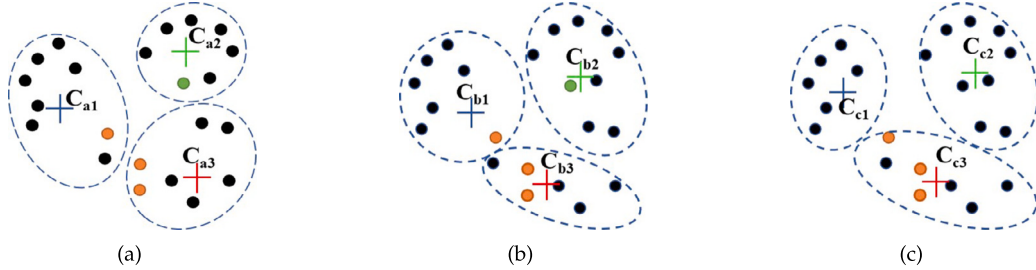


Fig. 4. The update processes of partition-based SSC. (a) Cluster centers (C_{a1} , C_{a2} and C_{a3}) are selected randomly firstly, each sample is allocated to the cluster where the nearest cluster center is located. (b) Cluster centers are updated to C_{b1} , C_{b2} and C_{b3} . (c) The above processes are updated iteratively under the guidance of label information until a threshold is met and the final clusters (C_{c1} , C_{c2} and C_{c3}) are obtained.

3. Research progress of semi-supervised clustering

In this section, various semi-supervised methods organized referring to the organization method of conventional clustering are analyzed and compared in detail. Suppose in dataset X , $X = \{x_1, x_2, \dots, x_i, \dots, x_n\}$, $x_i = \{x_{i1}, x_{i2}, \dots, x_{ij}, \dots, x_{im}\}$, x_{ij} denotes the j th attribute value for the i th data, n denotes the data size in X , m denotes the quantity of features (dimension). The class label set is $K = \{1, \dots, k, \dots, c\}$. The label of data x_i (denoted as L_{x_i}) is $L_{x_i} = k$, $k \in K$. The purpose of semi-supervised is to divide these points into groups based on several prior information. Taking Fig. 3 as an example of semi-supervised clustering process, the prior information is given in the form of class labels. Fig. 3(a) shows a two-dimensional dataset X , $X = \{x_1, x_2, \dots, x_9\}$, each sample contains two attributes. The labels of x_3 and x_4 are the same, and they are different from x_9 . This label information guides more accurate clustering results. In the sub-figure Fig. 3(b), $L_{x_1} = L_{x_2} = k_1$, $L_{x_3} = L_{x_4} = L_{x_5} = L_{x_6} = k_2$, $L_{x_7} = L_{x_8} = L_{x_9} = k_3$. As the number of clusters is not given, these points can be also grouped into four clusters as Fig. 3(c). At this time, $L_{x_1} = L_{x_2} = k_1$, $L_{x_3} = L_{x_4} = k_2$, $L_{x_5} = L_{x_6} = k_3$, $L_{x_7} = L_{x_8} = L_{x_9} = k_4$. The arbitrariness of the cluster shape and size in n -dimension data space is a great challenge in clustering process. When extended to semi-supervised clustering (SSC) using a limited number of prior knowledges, the challenge is also to predict a more accurate label of x_i .

3.1. Partition-based SSC

Partition-based semi-supervised clustering is usually improved on the traditional k-means and fuzzy c-means model, a small part is to improve the k-mode model for categorical data. The specific description will be detailed in this section. And the comparisons of these partition-based SSC are briefly shown in Table 1. Due to the large amount of literature involved, some methods are omitted from the comparison. Such as the constraint-kmeans [10] and CVQE, [32], most of the semi-supervised clustering are improvements on their ideology that make up for their weaknesses. Thus, constraint-based [10] and CVQE, [32] are not compared in Table 1.

The labeled data or pairwise constraints can guide the selection of cluster center and which cluster the samples belong to. When the target of clustering is expressed in terms of minimizing an objective function, partition-based SSC incorporates prior knowledge into the process of the calculation of optimal solution. An example of the process of partition-based SSC is presented in Fig. 4. The clusters are indicated by dotted line as boundary.

3.1.1. SSC based on conventional K-means model

K-means aims to partition each data object from the dataset to a unique cluster, it is a hard clustering method [9]. Its merit is that the calculation process is simple and efficient and it can cluster large-scale datasets. While its disadvantage is that it is sensitive to the selection of initial cluster center and the number of clusters. Based on traditional K-means, the K-means based SSC emerged. This type of method integrates prior information by the following three approaches to improve the clustering performance: (1) constraint-based methods [10], [133], [154], [157], [9], [32], [107], [60], and (2) distance-function learning methods [166], [175], [7], [143], [14] and (3) integrates the both two simultaneously [14], [130].

(1) Constraint-based methods initialize clustering and infer constraints based on the neighborhoods derived from labeled examples, enforce constraints during the clustering process or modify the clustering objective function. Basu et al. proposed

Table 1
Comparisons of partition-based SSC methods.

Method	Categories	Prior information	Time complexity	Evaluation index	Year	Advantages and disadvantages	Suitable scenario
Cop-Kmeans [133]	semi-supervised k-means	pairwise constraints	$O(nkt \cdot 2n)$	RI	2001	The method is easy to implement. While it is sensitive to order, the instance may not be assigned to a valid cluster if a wrong decision is made earlier.	The dimension is relatively small, and the value is small and continuous.
PCKmeans [9]	semi-supervised k-means	pairwise constraints	$O(nkt \cdot 2n + M + C)$	NMI, Pairwise F-measure	2004	It explicitly deals with pairwise constraint. While it can not handle noise constraints.	Suitable for large data sets and sparse high-dimensional data.
MPCKM [14]	semi-supervised k-means	pairwise constraints	$O(nkt \cdot 2n)$	Pairwise F-measure	2004	The performance is better than using only constraint-based or distance-based methods, however, it is poor on high-dimensional datasets.	Suitable for clusters with different shapes, and the data can be located in different subspace.
AFCC [44]	semi-supervised fuzzy c-means	pairwise constraints	$O(nkt + kt + nt)$	Ratio of well categorized points	2008	It partly solves the problem of determining an proper clusters number and proposes an active <i>SSFCM</i> using pairwise constraints.	It allows the number of constraints sufficiently low.
SSKFCM [166]	semi-supervised k-means	class labels	$O(nkt + n^2t)$	ACC	2009	It is time efficiency. However, for real applications, fuzzy membership degrees may not be available and difficult to generate from expert knowledge.	Suitable for clusters of any shape.
SCKMM [157]	semi-supervised k-means	pairwise constraints	$O(n'kt \cdot n^3 + M + C + k)$	RI, NMI	2010	It effectively solves the pairwise constraints violation, but the computational complexity is expensive and it can not handle noisy constraints.	Not suitable for large-scale problems.
ICop-Kmeans [154]	semi-supervised k-means	pairwise constraints	$O(nkt \cdot 2n)$	proportion of failure, F-measure	2012	It solves the order sensitive problem Cop-Kmeans, but it does not completely solve the problem of constraint violation.	Not applicable to large datasets.
SMUC [158]	semi-supervised fuzzy c-means	prior membership degree	$O(nkt \cdot n^3 + nt)$	NMI, ACC	2012	It avoids the choice on the parameter m. However, the average running times is longer.	Not applicable to large datasets.
LHC-S3FCM [40]	semi-supervised fuzzy c-means	class labels	$O(nkt + nT + 2knT)$	ACC	2018	The proposed graph-based regularization term can be easily embedded into the other objective function-based SSC methods, while the parameter m is difficult to estimate.	Not suitable to large datasets.
CS3FCM [39]	semi-supervised fuzzy c-means	class labels	$O(nkt + n^2 + nkT)$	ACC	2019	It estimates the safe confidence of each labeled sample. The parameter m is difficult to estimate.	Not suitable to large datasets.
$CrKMd - KNN/SVM/ANN/DT/RF$ [62]	semi-supervised k-modes	class labels	$O(nktk + nmt + kPnt + nm/n^3 / n/\log(n)/nm\log(n))$	MS, ARI, XB, %CP	2019	It uses credibilistic measure and machine learning techniques to handle the challenges of clustering categorical data. While it takes comparatively additional computation power and space.	Not applicable to uncertain data.
PCSKM [130]	semi-supervised k-means	pairwise constraints	$O(nkt \cdot 2nm + nmt + n_kkmt)$	Pairwise F-measure	2021	It considers the problem of data clustering with unidentified feature quality. While it can not effectively solve the pairwise constraints violation.	Suitable for large-scale and high-dimensional data.
SMKFC-ER [117]	semi-supervised k-means	class labels	$O(nkt + n^3tH + Ht)$	ACC, Precision, Recall, NMI	2021	It has the ability for adjusting cluster fuzziness and having closed-form solution. While it is time consuming.	Suitable for clusters of any shape.
novel Safe SSC [46]	semi-supervised fuzzy c-means	class labels	$O(nkt + n^2 + nkT + nT)$	ACC	2021	It safely uses the risk prior information, while the parameter m is difficult to estimate.	Not suitable to large datasets.
CSSC-Kmeans [60]	semi-supervised k-means	class labels	$O(t_1nmk_1 k + t_2nmk_2)$	NMI, RI	2022	The optimal cluster number and centroids is found through full use of label information, and it is not affected by cluster size and can adapt to clusters of arbitrary shape.	Suitable for large-scale datasets.

Nations:

$n/k, k_1, k_2/T, t, t_1, t_2$: the number of data set/clusters/iterations.

M/C: the must-link/cannot-link constraints set.

m/H/ n_k : the number of attribute value/mapping function/the kth cluster.

n' : the number of the examples after must-link constraints are merged.

P: the total number of attribute value.

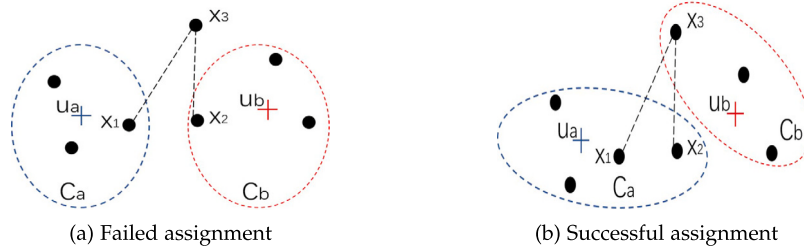


Fig. 5. The assignment of conflicts. (a) Object x_1 and x_2 are already allocated to the clusters C_a and C_b before deciding the tag of x_3 using Cop-Kmeans, as the existence of the cannot-link constraints related to x_3 , the method terminates. (b) Supposing $\text{dist}(x_3, u_b) < \text{dist}(x_3, u_a)$, x_3 is assigned to C_a and x_2 is temporarily removed from the cluster C_b when finding cannot-link constraints with x_2 and x_3 , then x_2 is reassigned to the closest cluster except the C_b .

seeded-kmeans and constrained-kmeans [10], seeded-kmeans only used the seeds information to initialize cluster centers, while the seeds labels in constrained-kmeans were kept unchanged during the iteration step. *Cop-Kmeans*, [133] is a typical constraint-based method, it takes pairwise constraints as background knowledge into the assignment step.

Based on the experiments in [133], the overall clustering accuracy (ACC), calculated by Rand index(RI), increased with the incorporation of constraints in each dataset. Compared with the k-means method with an accuracy of 58%, *Cop-Kmeans* had achieved significant improvement. When the quantity of pairwise constraints reached 100, in six datasets, the lowest overall accuracy up to 87%.

However, *Cop-Kmeans* has a deficiency of constraint violation, Fig. 5(a) shows one case of constraint violation in a dataset containing two clusters. To solve the above problem, *ICop-Kmeans*, [154], an improved *Cop-Kmeans*, was proposed. *ICop-Kmeans* handled this situation by choosing the cluster with the smallest distance between x_3 to the cluster center u_a and u_b (as Fig. 5(b) shows).

The experiments in [154] demonstrated *ICop-Kmeans* made great success in the statistics of failure proportion against *Cop-Kmeans*, their *F-measure* was almost the equivalent. When the number of the given pairwise constraints increased, *ICop-Kmeans* had a better time efficiency than *Cop-Kmeans*, because the *Cop-Kmeans* frequently restarted and selected the cluster centroids at random initial.

COP-KMeans and *ICop-KMeans* not handles soft constraints (constraints that can be violated with an associated violation cost), they fail to explicitly deal with pairwise constraints. *CVQE*, [32] and *LCVQE*, [107] allow some constraints unsatisfied (soft constraints). Besides, the violations were also permitted as “soft” in *PCKmeans*, [9]. *PCKmeans* expresses the clustering target in terms of minimizing an objective function: the first term denotes aggregate distance among the dates and cluster centers, the second term is expense of violation of pairwise constraints (as shown in eq. (1)).

$$\min J_{PCKmeans} = \frac{1}{2} \sum_{x_i \in X} \|x_i - \mu_{L_{x_i}}\|^2 + \sum_{x_i, x_{i'} \in M} w_{x_i, x_{i'}} \mathbb{I}(L_{x_i} \neq L_{x_{i'}}) + \sum_{x_i, x_{i'} \in C} \overline{w}_{x_i, x_{i'}} \mathbb{I}(L_{x_i} = L_{x_{i'}}) \quad (1)$$

where $\mu_{L_{x_i}}$ means the L_{x_i} th center prototype selected by the provided pairwise constraints, M , C represents must-link, cannot-link set respectively, $W = \{w_{x_i, x_{i'}}\}$ and $\overline{W} = \{\overline{w}_{x_i, x_{i'}}\}$ indicate the weights, $\mathbb{I}$ indicates the function of indicator and $\mathbb{I}[true] = 1$, $\mathbb{I}[false] = 0$. The meanings of these notations are not repeated when they appear in the following equations.

Based on the experiment in [9], the value of *NMI* (Normalized Mutual Information) increases with the pairwise constraints quantity growing. On the dataset of *News-diff3*, the value of *NMI* up to 0.85, it demonstrates *PCKmeans* performs considerably better compared with unsupervised clustering with zero pairwise constraints. The scores on *NMI* and *F-measure* grow and the performance of objective function improves with the pairwise constraints number increasing.

To find the optimal cluster centroids and the number of clusters, a compact-cluster-based semi-supervised Kmeans-like clustering method, named as *CSSC-Kmeans*, [60], is proposed. In this method, an item is added to the objective function of traditional K-means to measure the compact degree of clusters.

Constraint-based semi-supervised k-means clustering method is easy to implement, and it gradually and effectively solves the problem of constraint violation. While the distance between data generally uses Euclidean distance, and the data may not be well separated. Distance-function learning methods can effectively solve this problem.

(2) Distance-function learning methods use prior information to adapt the distance metric. Since the data with original features may not be designated in a space in which clusters can be adequately isolated, it is necessary to modify the distance metric to warp the space such that the distances among same/different cluster are minimize/maximize. Using symmetric positive-definite matrix A to parameterize Euclidean distance is a distance-function learning method. The same parameterization was applied previously by [7], [143]. Many approaches treat metric learning as a convex optimization problem, Xing et al. [143] got an adaptive distance measure based on the given pairwise constraints. The objective function is shown in eq. (2).

$$\begin{cases} \min J_A = \sum_{x_i, x_{i'} \in M} \|x_i - x_{i'}\|_A^2 - \log(\sum_{x_i, x_{i'} \in C} \|x_i - x_{i'}\|_A) \\ s.t. A \geq 0 \end{cases} \quad (2)$$

where $(\|x_i - x_{i'}\|_A = \sqrt{(x_i - x_{i'})^T A (x_i - x_{i'})})$, the second term is used to ensure A does not collapse the dataset into a single point.

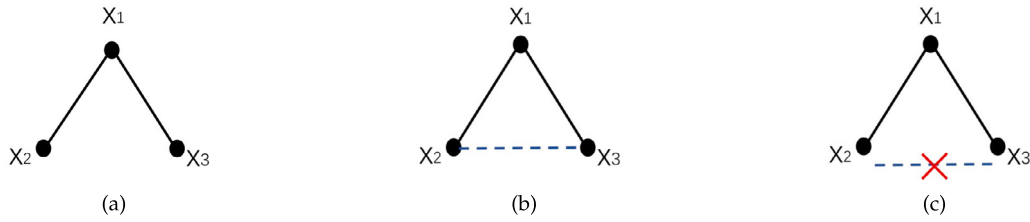


Fig. 6. The method to deal with constraint violation. (a) There are must-link constraints between x_1 and x_2 , x_1 and x_3 , in view of the transitivity of the must-link constraints, x_2 and x_3 should have must-link constraint. (b) The conflict occurs as a cannot-link constraint exists between x_2 and x_3 . (c) Removing the constraints with constraint violation.

Similarly, *MK-MEANS*, [14] used a separate weight matrix $A_{L_{x_i}}$ for each cluster, the second term of *MK-MEANS* was generated from the normalizing constant of the L_{x_i} -th Gaussian with covariance matrix $A_{L_{x_i}}^{-1}$. Based on the experiences in [14], on the *Wine* datasets, the difference *F-measure* between *MK-MEANS* (utilize metric learning) and *PCKmeans* (do not utilize metric learning) up to 0.3. Even in the absence of constraints, *MK-MEANS* also improves clustering accuracy. However, when the matrix A is not diagonal, this method is very expensive.

Distance-function learning *SSC* methods only incorporate prior information to the distance metric. To achieve better clustering effect, some researches integrate constraint-based and distance-function learning simultaneously into a unified and principled framework.

(3) Based on the techniques of constraint-based and distance-function learning, *MPCKM*, [14] is proposed. The objective function of *MPCKM* is:

$$\begin{aligned} \min J_{MPCKM} = & \sum_{x_i \in X} (\|x_i - \mu_{L_{x_i}}\|_{A_{L_{x_i}}}^2 - \log(\det(A_{L_{x_i}}))) + \sum_{(x_i, x_{i'}) \in M} w_{x_i, x_{i'}} f_M(x_i, x_{i'}) \mathbb{I}(L_{x_i} \neq L_{x_{i'}}) \\ & + \sum_{(x_i, x_{i'}) \in C} \overline{w_{x_i, x_{i'}}} f_C(x_i, x_{i'}) \mathbb{I}(L_{x_i} = L_{x_{i'}}) \end{aligned} \quad (3)$$

MPCKM used $A_{L_{x_i}}$ for cluster L_{x_i} , which permitted clusters of different shapes. Functions $f_M(x_i, x_{i'})$ and $f_C(x_i, x_{i'})$ aimed to make all constraint violations treated unequally. However, to solve the pairwise constraint violation issue, *MPCKM* was not enough effectively. Similarly, the *ASSKMM*, [145] added the cost term to the objective in the feature space. To arrive at the goal, [157] incorporated the must-link constraints into the corresponding closures to form the new cannot-link constraints. In view of the constraint violation in Fig. 6(b), a new adaptive *SSC* method, called *SCKMM*, was presented in [157] based on kernel metric learning. Before clustering, *SCKMM* added an inspection step to remove those cannot-link constraints directly (as Fig. 6(c) shows) to ensure the transitivity is not violated. *SCKMM* is more effective to solve the constraints violation issue.

On the basis of *MPCKM* method and the work of on sparse K-Means clustering, *PCSKM*, [130] was proposed. The objective function of this method (as shown in eq. (4)) was modified by incorporating constraints to *MPCKM* to select the needed subgroups of features for clustering. Its strategies in clustering and active learning can be easily scalable to large or high dimensional datasets.

$$\begin{cases} \min J_{PCSKM} = \sum_{j=1}^m w_j \left[\sum_{i=1}^n (x_{ij} - \eta_{1j})^2 \right] - \left(\sum_{c=1}^k \sum_{(i=1, x_i \in c_k)} (x_{ij} - \mu_{1j})^2 \right) + \sum_{(x_i, x_{i'}) \in M} f_M(x_i, x_{i'}) \mathbb{I}(L_{x_i} \neq L_{x_{i'}}) \\ \quad + \sum_{(x_i, x_{i'}) \in C} f_C(x_i, x_{i'}) \mathbb{I}(L_{x_i} = L_{x_{i'}}) \\ \text{s.t.} \quad \sum_{j=1}^m w_j^2 \leq 1, \sum_{j=1}^m |w_j| \leq s, w_j \geq 0 \forall j \end{cases} \quad (4)$$

In eq. (4), $\eta_1 = [\eta_{11}, \eta_{12}, \dots, \eta_{1m}]$ is the average of the whole data (global center). Although the purpose of *PCSKM* was to select an appropriate feature representation rather than its performance, the experiments in [130] showed that *PCSKM* had better performance. Compared with *PCKmeans* and *MPCKM*, the *F-measure* value of *PCSKM* is higher, except in some cases when the Seeding method [14] is used for initialization. Even in the case of poor performance, the difference from the best method is tiny little. For *MPCKM*, its performance is better than *PCKmeans* and *MK-MEANS*, as *MPCKM* combines the constraint-based and distance-based method.

The integration of constraint-based and distance-based semi-supervised k-means clustering can solve the constraints violation more effectively, while it still continues the hard clustering feature of K-means. To generate a fuzzy partition (give each object a membership degree in each cluster), *SSC* based on fuzzy c-means is proposed to improve *SSC* based on k-means.

3.1.2. *SSC* based on fuzzy c-means model

Compared with the hard clustering results generated by k-means, *FCM* provides more flexible clustering results, it assigns a membership level to each object in each cluster. *SSFCM*, [117] modifies the objective function of *FCM* by incorporating the supervising information. The proposed *SSFCM* can be formulated by the objective function of:

$$\min J_{SSFCM} = \sum_{k=1}^c \sum_{i=1}^n u_{ik}^m d_{ik}^2 + \alpha \times \sum_{k=1}^c \sum_{i=1}^n (u_{ik} - \bar{u}_{ik})^m d_{ik}^2 \quad (5)$$

In the above equation, m is bigger than 1 and it represents the fuzzy degree, u_{ik} denotes the degree of membership of the i th samples to the k th cluster, d_{ik} means the distance among the i th data and the k th center prototype, α is a coefficient that is non-negative and represents the importance degree of supervision, $\bar{u}_{ik}$ denotes the degree of membership of tagged data.

There are several caveats of the *SSFCM* method [166], [117].

1. The relationship between instances is generally reflected applying the Euclidean distance in most of the methods. While in real-world application, when each attribute of the sample is independent, Euclidean distance works poorly.

2. The choice of parameter m is difficult.

3. The inconsistency between the obtained membership degrees and the given prior knowledge generally minimized by geometric metrics. While it should make greater use of physically meaningful and conceptual measures to replace the geometrical measures.

For improving the performance of *SSFCM*, many improved methods are proposed, and these methods are displayed in three types:

(1) Distance learning methods [61], [166], [117], [158]: The relationship between instances is generally reflected applying the Euclidean distance in *SSFCM*. While the pairwise constraints can be utilized to optimize the distance, such as in [61], they first elevated the pairwise constraints from point level to group level, then learnt a novel distance learning method from the group-level information. In real-world application, when each attribute of the sample is independent, Euclidean distance works poorly. [158] focused on learning a Mahalanobis distance metric from prior information given by the user to displace the Euclidean distance in *FCM*-based methods. [166] used kernel methods to obtain the best clusters (called *SSKFCM*), it balanced the contribution of unsupervised and supervised part in a novel defined objective function eq. (6).

$$\min J_{SSKFCM} = W_L \sum_{i=1}^L \sum_{k=1}^c (u_{ik}^m - \bar{u}_{ik}^m)(K(y_i, \bar{\mu}_k) - K(y_i, \mu_k)) + W_U \sum_{i=1}^U \sum_{k=1}^c u_{ik}^m (2 - 2K(x_i, \mu_k)) \quad (6)$$

W_L and W_U are the weight for unsupervised and supervised part, and the sum of W_L and W_U is 1. For a given unlabeled data set $\mathbb{U}$, $\mathbb{U} = \{x_1, x_2, \dots, x_r, \dots, x_U\}$, and a labeled data set $\mathbb{L}$, $\mathbb{L} = \{y_1, y_2, \dots, y_l, \dots, y_L\}$, in the input data space of d -dimensional, a mapping function ϕ transforms the data to a higher dimensional feature space. $K(x_i, x_{i'})$ is a kernel function and it is given by $K(x_i, x_{i'}) = \phi(x_i) \cdot \phi(x_{i'})$, $\phi(x_i)$ and $\phi(x_{i'})$ are the images of x_i and $x_{i'}$ in the feature space.

In *SSKFCM*, it can identify clusters with the desired shapes as using multiple kernels to learn non-linear distances. As *SSFCM* has no data mapping, *SSKFCM* has better performance than *SSFCM* based on the experiment in [166]. The quantity of tagged data affects the performance of *SSKFCM*, the clustering accuracy up to 100 after increasing the quantity of tagged data on the artificial dataset of *circles* and real-world dataset *Iris*. Compared with other kernel-based SSC methods, one of the advantages of *SSKFCM* is time efficiency. It is because that *SSKFCM* transform the objective function adopting a deflection technique, and it enables the optimal solutions unchanged.

While for single kernel learning, choosing a proper kernel function is essential. To solve the difficult issue of selecting kernel function, a semi-supervised fuzzy clustering was proposed in [117] as shown in eq. (7). The performance of these methods has been improved, while reducing the time complexity should be considered to the next step.

(2) The inconsistency learning methods between the obtained membership degrees and the given prior knowledge [117], [51], [158]. The inconsistency is generally minimized by geometric metrics. While it should make greater use of physically meaningful and conceptual measures to replace the geometrical measures. [51] proposed a novel *FSREC-PC* method, it introduced the entropy regularization into the objective function. [117] proposed semi-supervised multiple kernel fuzzy clustering based on entropy and relative entropy called *SMKFC-ER*. The relative entropy divergence metric was used to replace the geometric distance metric for semi-supervised process, as shown in the third term of eq. (7). In addition, the choice of parameter m is difficult, *SMKFC-ER* also settled the parameter problem. *SMKFC-ER* applied entropy coefficient, the objective function's second term, instead of fuzzifier m for unsupervised process.

$$\min J_{SMKFC-ER} = \sum_{k=1}^c \sum_{i=1}^n u_{ik} D_{ik}^2 + \gamma \sum_{k=1}^c \sum_{i=1}^n u_{ik} \ln u_{ik} + \beta \times \ell_i \sum_{k=1}^c \sum_{i=1}^n u_{ik} \ln \left(\frac{u_{ik}}{\bar{u}_{ik}} \right) + \delta^{-1} \sum_{h=1}^H \omega_h \ln \omega_h \quad (7)$$

In eq. (7), γ determines the degree of fuzziness bigger than 0, β determines the importance degree of the semi-supervised composition, δ indicates the coefficient for entropy of kernel weights. $D_{ik}^2 = \|\Omega(x_n), \bar{\mu}_k\|$ shows the distance induced by the kernel from data samples and the center, where $\Omega(x) = \sum_{h=1}^H \sqrt{\omega_h} \phi_h(x)$, ω_h is the weight of the h th kernel function, ℓ_i with 1 and 0 determines whether the data is labeled. Its ability of identifying non-spherical clusters was enhanced by extending to a multi-kernel version. $\delta > 0$ is the entropy coefficient of kernel weights.

Based on the experiment in [117], the proposed method was already demonstrated its superiority over the comparative methods on real-world and synthetic datasets, such as *FCM*, [137], *SSFCM*, [106], *SFC-ER*, [117]. The above-mentioned methods assume the association among the labeled data and the final clustering performance is always positive, while in real life they may destroy the performance of semi-supervised clustering.

A newly developed metric-based *SSFCM* called *SMUC* was given in [158]. It is a prior membership degree based *SSFCM*. When the given information is considered as the preset membership degree of the data, the distinction among the fuzzy membership degree with the pre-given fuzzy membership degree is integrating into the *FCM* objective function. The objective function (as shown in eq. (8)) introduced metric learning and entropy regularization to the traditional *FCM*, so that the above-mentioned two question could be solved simultaneously.

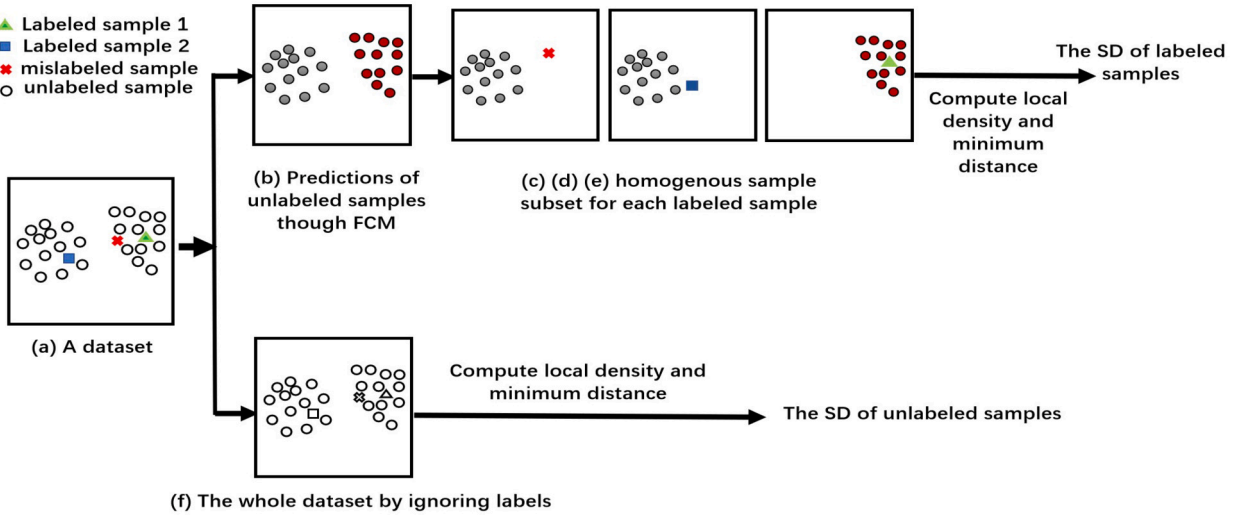


Fig. 7. The process of estimating SD of labeled and unlabeled data.

$$\min J_{SMUC} = \sum_{k=1}^c \sum_{i=1}^n u_{ik} \|x_i - \mu_k\|_A^2 + \eta^{-1} \sum_{k=1}^c \sum_{i=1}^n (u_{ik} - \bar{u}_{ik}) \ln(u_{ik} - \bar{u}_{ik}) \quad (8)$$

In [158], *SMUC* is compared with *sFCM*, [156], *eFCM*, [156], *SSKFCM*, *AFCC*, [44] etc., *sFCM* just replaces u_{ik}^m with $(u_{ik} - \bar{u}_{ik})^m$ in objective function of *FCM*, both *eFCM* and *SMUC* generate cluster membership degrees applying maximum entropy. As the learned distance metric can force the samples in different clusters to be far away and in one cluster as close as possible, *SMUC* is superior to *eFCM* on every dataset. *SMUC* realizes the greatest performance except on the *WDBC* dataset. Since it is considerably effective for *SMUC* to employ the tagged and untagged samples simultaneously to enhance the performance, *SMUC* is comparable to *AFCC* in these cases. However, the average running times of *SMUC* is longer than these methods, the average running time of *SMUC* on the dataset of *Vowel* is 18.78s while *FCM* need only 2.56s.

(3) Others [40], [39], [46], [44]: In view of the deficiency of labeled data may destroy the performance of semi-supervised clustering, some studies focus on Safe Semi-supervised Learning (*S3L*) [39]. As the sample labels assigned by the human may be different from their true labels, it is vital to assign different safe confidences to different data points. Under the basis of this assumption, Gan et al. proposed two safe *SSC* methods, *LHC-S3FCM*, [40] and *CS3FCM*, [39] (as shown in eq. (9) and eq. (10)). Guo et al. [46] proposed another safe semi-supervised clustering method to explore the labeled and unlabeled samples simultaneously. All of them restricted the outputs of the tagged samples by using the local graph. While the difference is *CS3FCM* adds confidence weight, [46] first computes the safe degree (*SD*) (shown in Fig. 7).

$$\min J_{LHC-S3FCM} = \sum_{k=1}^c \sum_{i=1}^n u_{ik}^m d_{ik}^2 + \lambda_1 \sum_{k=1}^c \sum_{i=1}^L (u_{ik} - \bar{u}_{ik} \ell_i)^m d_{ik}^2 + \lambda_2 \sum_{i=1}^L \sum_{r=1}^N W_{ir} \sum_{k=1}^c (u_{ik} - u_{rk})^2 \quad (9)$$

$$\min J_{CS3FCM} = \sum_{k=1}^c \sum_{i=1}^n u_{ik}^m d_{ik}^2 + \lambda_1 \sum_{i=1}^L S_k \sum_{k=1}^c (u_{ik} - \bar{u}_{ik} \ell_i)^m d_{ik}^2 + \lambda_2 \sum_{i=1}^L \frac{1}{S_k} \sum_{r=1}^N W_{ir} \sum_{k=1}^c (u_{ik} - u_{rk})^2 \quad (10)$$

$$\min J_C = \sum_{r=1}^U \sum_{k=1}^c \tilde{\gamma}_r u_{rk}^m d_{rk}^2 + \lambda_1 \sum_{i=1}^L \tilde{\gamma}_i \sum_{k=1}^c (u_{ik} - \bar{u}_{ik} \ell_i)^2 d_{ik}^2 + \lambda_2 \sum_{i=1}^L (1 - \tilde{\gamma}_i) \sum_{r=1}^N W_{ir} \sum_{k=1}^c (u_{ik} - u_{rk})^2 \quad (11)$$

λ_1 and λ_2 are the parameters of regularization, W_{ir} is the connection among the tagged and untagged data, when the unlabeled data x_i is the p nearest neighbors of the labeled data x_r and they have same label obtained from *FCM*, $W_{ir} = 1$, otherwise, 0, S_k is the safe confidence. γ_r and γ_i denote the obtained *SD* of labeled and unlabeled data. Among the objective function of eq. (9), eq. (10) and eq. (11), the term second and third respectively limit the consistency of clustering results with given class labels and the homogeneous neighbors.

To obtain the *SD* of labeled sample, (1) the cluster predictions are obtained by *FCM*, (2) the cluster predictions are mapped to the equivalent given labels by using Kuhn-Munkres method [174] to get the homogenous sample subsets, (3) the *SD* of labeled sample is obtained by evaluating the density of localization and the distance of minimum. To obtain the *SD* of unlabeled sample, local density and minimum distance is computed on the whole dataset by ignoring labels. Then the graph-based regularization term is constructed in eq. (11) to make the outputs of risky labeled samples the same as their nearest unlabeled neighbors.

The accuracy of the different clustering methods *LHC-S3FCM* and *CS3FCM* are shown in [39]. With the wrong ratios of prior knowledge changing, *SSFCM* and *CS3FCM* outperform *FCM* in most cases. As the wrong ratios increase, the overall performance of *SSFCM*, *LHC-S3FCM* and *CS3FCM* decreases on most comparison datasets. These experiments result also indicates the performance of semi-supervised clustering degrades as the exist of the wrong labels. Based on the experiments in [46]. Compared with *SSFCM*

and FCM on ten UCI real datasets, [46] obtains the best performance. On *Australian*, *Balance-scale*, *Wine*, *Tae* and *Heart* datasets, it achieves better performance than *LHC-S3FCM* and *CS3FCM*. It is demonstrated that this proposed method is effective and the obtained labeled samples are applied safely.

In view of finding an appropriate number of clusters in *SSFCM*-based methods is difficult, [44] proposed a pairwise constraints based semi-supervised fuzzy clustering, it added regularization term and pairwise constraints cost term to the objective function. The objective function (as shown in eq. (12)) - *AFCC* - extended the *PCKmeans* method to fuzzy clustering.

$$\min J_{AFCC} = \sum_{k=1}^c \sum_{l=1}^n u_{ik}^m d_{ik}^2 + \alpha \times \left(\sum_{((x_i, x_{i'})) \in M} \sum_{k=1}^c \sum_{l=1, l \neq k}^c u_{ik} u_{i'l} + \sum_{((x_i, x_{i'})) \in C} \sum_{k=1}^c u_{ik} u_{i'l} - \beta \sum_{k=1}^c \left[\sum_{l=1}^n (u_{ik}) \right]^2 \right) \quad (12)$$

In eq. (12), α designates the significance of the supervision term, β denotes the regularization parameter.

In contrast to *PCKmeans*, the ratio of well categorized points is increased from 0.7 to 0.97 on the *Arabidopsis* database. On the *Iris* data set, its performance is also improved although the ratio of well categorized points only increases 0.1. However, the time-consuming would be expensive.

In view of the missing of the data attributes, [41] proposed a semi-supervised fuzzy c-means clustering algorithm *SSFCM-ID* for incomplete data. First, the linear interpolation imputation technique [103] is used to initially fill the missing features in both labeled and unlabeled data. Then the labeled data is employed to initial the cluster centers and the fuzzy partition matrix is initialed randomly. Finally, the algorithm iterates until the abort condition is met. The experiments in [103] have demonstrated that the proposed methods can handle incomplete data efficiently. While with the missing ratio increasing more than %15, the proposed algorithm does not give better results.

SSC based on k-means and fuzzy k-means have been greatly improved in processing numerical data, which can be used to process a large number of high-dimensional data, however, it is not suitable for processing categorical data [97]. K-modes method is a non-numerical attribute version of K-means method [62], it uses the measure of dissimilarity matching for categorical data.

3.1.3. SSC based on fuzzy k-modes [62]

When the attribute values of the sample are the same as the cluster center in K-modes, the distance is 0, otherwise it is 1. Fuzzy k-modes minimizes eq. (13) to divide the dataset X in K clusters.

$$\min J_{fuzzy\ k-mode} = \sum_{i=1}^n \sum_{k=1}^c u_{ik}^\eta D(c_k, x_i) \quad (13)$$

where η denotes the fuzzy exponent. $u = [u_{ik}]$ is a $c \times n$ matrix, u_{ik} is the membership degree. In the start of fuzzy k-modes method, K modes are randomly selected as initialization.

The semi-supervised technique in [62] first using the proposed Credibilistic *K-Mode* to partition the dataset into k clusters. Then the obtained k clusters were used to construct a supervised model (called *MLCrKmd*), after that *MLCrKmd* classified other unlabeled or uncertain samples. It is reported that *CrKmd* outperforms the comparison method for all the datasets mentioned in [62], however, the unique cluster of the data points cannot be determined in *CrKmd* as multiple clusters have the same credibility. To handle this issue, various machine learning techniques (*CrKmd-KNN*, *CrK-Md-SVM*, *CrKmd-DT*, *CrKmd-ANN* and *CrKmd-RF*) are integrated with *CrKmd*.

According to several cluster validity indices (Minkowski Score (*MS*), Adjusted Rand Index (*ARI*), Xie–Beni index (*XB*) and Percentage of Correct Pair (*%CP*)) computed in [62], one technique known as *MLCrKmd* integrating machine learning and *CrKmd* produces the best result. They use credibilistic measure and machine learning techniques to handle the challenges of clustering categorical data. While they take comparatively additional computation power and space, and not applicable to uncertain data.

For the partition-based SSC, many experiments have demonstrated the clustering performance can be improved by adapting the similarity measure (distance learning-based) or modifying the search for appropriate clusters (constraints-based) to add the supervising information to the clustering process, and the degree of improvement is almost the same, such as in [14]. Besides, as partition-based SSC finds the disjoint k partitioning of a dataset by minimizing an objective function, it is also easy to realize both constraints-based and distance learning-based SSC. While the time complexity of some distance learning-based methods is extremely high. Therefore, most of the partition-based SSC methods are not applicable to large and high-dimensional datasets. For the improved methods of semi-supervised fuzzy c-means, some methods can recognize clusters of arbitrary shapes, but the improved semi-supervised k-means cannot.

3.2. Hierarchical-based SSC

Most semi-supervised hierarchical clustering methods (as Table 2 shows) are the variant of the single link, complete link and average link method, they are different in the method of describing the similarity between two clusters. Their similarity calculating is depicted in Fig. 8. The single/complete link distance selects the smallest/largest value among the distances of all sample pairs. While in the average link distance, the two clusters are partitioned by the average distances among all samples in pairs. In addition to the above three linkage methods, there are two other linkage methods (Ward's Linkage and Centroid Linkage [127]) in traditional unsupervised hierarchical clustering. Ward's Linkage aims at minimizing the total within-cluster variance, it is less susceptible to noise interference but it is only applicable to Euclidean distance. Centroid Linkage calculates the distance between the center points,

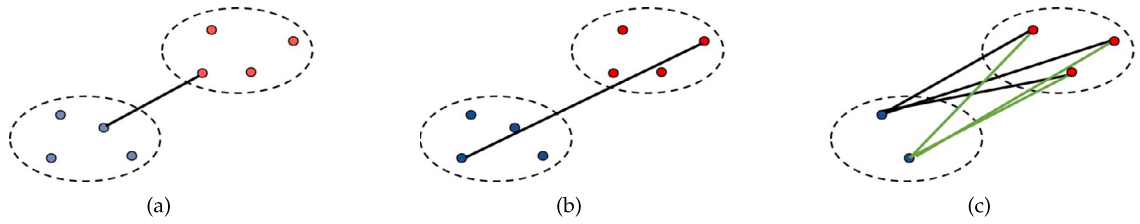


Fig. 8. The distance of (a) single link method, (b) complete link method and (c) average link method.

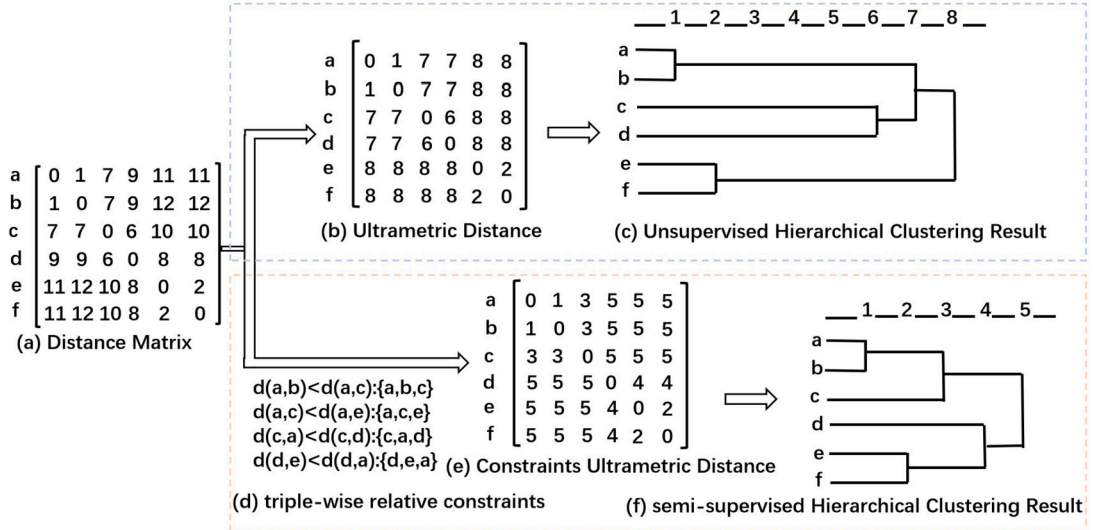


Fig. 9. An example illustrates hierarchical-based SSC. The process of (a) to (b) represents the original dissimilarity matrix is transformed into an ultra-metric distance matrix. (d) gives four triple-wise relative constraints, in which $d(c,a) < d(c,d) : \{c,a,d\}$ means the distance from c to a is smaller than the distance from c to d. The process of (a) to (e) represents the transformation under the given triple-wise relative constraints.

it is less used as the distance is not monotonic during the iteration. Therefore, this section only summarizes and analyzes the semi-supervised hierarchical clustering methods using single link, complete link and average link method.

The bottom-top hierarchical clustering establishes a tree structure by finding the two clusters with the shortest distance at a time, after that merges the clusters into a large cluster until they are all merged into one cluster. As soon as the above clustering tree is obtained, the number of clusters can be obtained directly according to the tree structure. When the amount of clusters changes, it is not necessary to establish a new tree structure again. However, in the process of calculating distance and merging clusters, there may be some deviation between the obtained results and the real results. In order to reduce the deviation and obtain more accurate clustering results, many semi-supervised hierarchical clustering methods are proposed.

(1) For the single link-based semi-supervised hierarchical clustering methods [171], [31], [98], [16], Sadaaki et al. [98] introduced pairwise constraints to the single linkage directly. Although this method can handle must-link and cannot-link simply, these constraints should be consistent. [30] tried to use a centroid distance function to reduce the average complexity of computing distances between two clusters. The *IPoptim* (based on optimization) and *UltraTran* (based on the transitive dissimilarity) [171] were two proposed hierarchical-based SSC frameworks. The first method *IPoptim* constructed a matrix of optimal distance that simultaneously satisfying the ultra-metric and relative constraints. *UltraTran* constructed the transitive dissimilarity by incorporating the relative constraints into its process.

An example illustrating hierarchical-based SSC with triple-wise relative constraints is displayed in Fig. 9. Take the first layer merge in the process of unsupervised clustering as an example, the distance between sample c and d is 6, which is less than all the distances between d or c are revised to the minimum distance value among them. In the process of the SSC, because the existence of constraints $d(c,a) < d(c,d) : \{c,a,d\}$, sample c and d fail to merge, therefore, the ultra-metric distance of c to d is not available in first level merge.

The experiment in [171] observed the efficiency of *IPoptim* and *UltraTran* will not increase monotonically with the increase of constraints, because that their goal was not to meet all constraints, but to seek out a kind of efficient approximation of the constrained ultra-metric. Compared with *HAC* and *HACoc*, [168], *HACoc* takes much longer to run especially on large datasets, although it obtains the optimal performance. As the constraints amount grows, the performance of *IPoptim* and *UltraTran* enhances a lot.

Table 2
Comparisons of hierarchical-based SSC methods.

Method	Prior information	Time complexity	Evaluation index	Year	Advantages and disadvantages	Suitable scenario
COP-COBWEB [132]	pairwise constraints	$O(n \log n)$	ACC	2000	It is the first method to use instance-level hard constraints to improve the performance of clustering.	Not suitable to large-scale data sets.
CCL [66]	pairwise constraints	$O(n^2 \log n)$	CRI	2002	It is also robust when over-constrained, and it additionally proposes an active learning method to increase the value of constraints.	Suitable to detect clusters are not separated in the feature space.
Learning Threshold method [29]	class labels	$O(n^2)$	F-measure	2006	The proposed solution is applicable to any form of hierarchical clustering. While it is affected by class distribution.	Not suitable to the sparseness data.
Agglomerative Clustering with Constraints [31]	must-link, cannot-link, δ -constraint, ϵ -constraint, γ -constraint	$O(n)$	Distortion, Purity	2010	It introduces the new constraint, and reduces computational time.	Not suitable to the important asymmetric distance situation.
IPoptim [171]	triple-wise relative constraints	$O(n^2)$	F-measure	2011	It considers both ultra-metric and triple-wise relative constraints in the process of optimization. While the performance is not monotonically increasing with the constraints numbers.	Suitable to large datasets.
UltraTran [171]	triple-wise relative constraints	$O(n^3)$	F-measure	2011	It modifies the Floyd-Warshall method, the updated value of minimal transitive dissimilarity is limited by any constraints that specify its merge orders. While the performance is not monotonically increasing with the constraints numbers.	Suitable to large datasets.
HCAC [16]	cluster-level constraints	$O(n)$	F-measure	2016	It is specially efficient when dealing with more than two clusters. The cluster-level constraints reduce the number of pairs to be analyzed by the user.	Suitable to datasets with clusters that contain boundaries that are hard to detect.
SCECH [141]	class labels, pairwise constraints	$O(n \log n)$	F-measure	2016	It obtains higher accuracy as combining CECH and two supervising information (class labels and pairwise constraints), and the performance is stable.	Suitable for clusters of any shape.
Semi-MultiCons [149]	pairwise constraints	$O(n^2)$	NMI, ACC, Purity	2022	It attempts to correctly infer the number of clusters, and can process the data imbalanced data. Besides, the number of clusters is large, different, ambiguous or unknown in the base clustering members can be also processed.	Suitable for datasets with large number of instances.

[31] used the instance-level (must-link and cannot-link) and cluster-level constraints (δ -constraint represents two points located in various clusters are at a minimum δ away from each other, ϵ -constraint represents two points located in one clusters are at most ϵ away from each other) to perform agglomerative hierarchical clustering.

The value of distortion improvement for constrained clustering is of the order of 15% according to the experiments in [31]. And the average percentage purity for constrained clustering also improves, one possible reason is that the must-link constraints boot method for better cluster when many pairs of clusters have similar short distances in the bottom of the dendrogram, conversely, the cannot-link constraints exclude unwise connections in the top of the dendrogram.

Furthermore, to considerably enhance the efficiency of agglomerative clustering, the new γ constraint (γ constraint requires a pair of clusters whose geometric centers must be smaller than γ) is introduced. As the usage of the γ -constraint and its combination with the triangle inequality, the proposed methods in [31] can be applied to any metric space including single and complete link clustering.

By simultaneously utilizing the triangle inequality and γ constraint, the overall amount of calculation saves about 10% according to the experiments in [31], the average clustering percentage purity also improves.

However, the above methods only consider whether to merge the pair of elements according to the distance. When the pair of elements represent different concepts, they would be the first to be merged as they are the nearest. To take this situation into account, HCAC [16] was proposed, it also used cluster-level constraints.

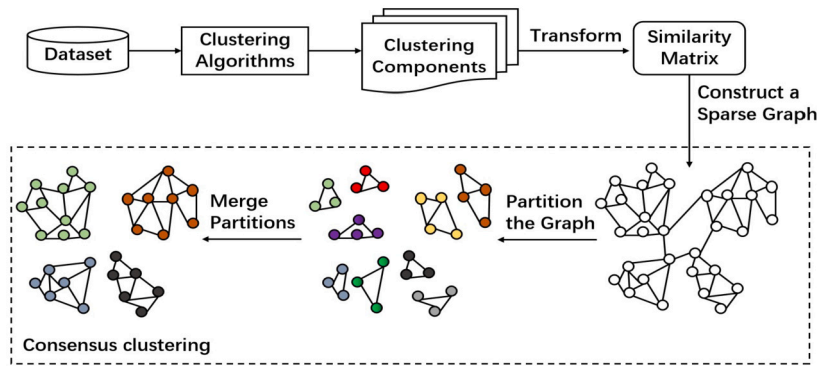


Fig. 10. The framework of CECH.

(2) For the complete link-based semi-supervised hierarchical clustering methods [66], [29], [31], [33], *CCL*, [66] was proposed by distorting instance-level constraints to reflect the spatial information. It can successfully incorporate constraints into different types of datasets. [29] is another proposed complete link-based semi-supervised hierarchical clustering method. As the actual final partition of hierarchical clustering depends on a threshold placed on the similarity measure, this method utilizes a small number of labeled instances and the F-measure for complete-link hierarchical agglomerative clustering. Constraints were used directly to the clustering process in most approaches, while this approach used the labeled data to find a good threshold value, then the threshold value was utilized when clustering.

Although the above methods achieve the improvement of clustering performance, the instance-level constraints they used can be improved to cluster-level constraints [31], [33] to reduce the problem that the relationship between clusters is affected by the constraints between instances.

(3) For the average link-based semi-supervised hierarchical clustering methods [132], [141], [132] was proposed to show how the instance constraints can be incorporated into a clustering method. It is the first method to propose two types of instance-level constraints (must-link and cannot-link constraints). *SCECH*, [141] was improved by *CECH*, [140], in which pairs of clusters were merged when they share relatively strong inter connectivity. In *SCECH*, it first used the framework of *CECH* (shown in Fig. 10) to construct a sparse graph based on the similarity matrix obtained by multiple clustering components. Second, the graph was partitioned into sub-clusters. And finally, the partitions were merged. In the second step, *SCECH* considered both the cases of class label and pairwise constraints. In the case of obtaining label information, the tagged data that fall into one class in the sparse graph were considered as the same sub-cluster, and the rest graph was partitioned using *METIS*, [63] (excluding tagged data). While in the case of pairwise constraints, this method focuses on modifying *SCECH* with cannot-link constraints. If the sub-cluster includes cannot-link constraints, it will be partitioned continuously.

Experiments in [141] show the *F-measure* value compared with other three semi-supervised ensemble methods (*SCEV*, [85], *SS-CSPA*, [93] and *SSCC*, [56]). From its statistics, *SCECH* performs better. It also shows that all methods enhance the performance as the percentage of pairwise constraints grow, specifically for constraints of 0%, which implies that *SCECH* degenerates to *CECH*.

In contrast to these hierarchical-based *SSC* methods described above, a recently proposed method –*Semi-MultiCons*, [149]– uses the idea of ensemble for semi-supervised clustering. First, the closed patterns can be extracted based on the transformation of the base clustering members. Then the *Semi-MultiCons* using the idea of hierarchical clustering is employed to process these closed patterns according to supervising information. Finally, the clustering is identified. Compared with a single clustering, *Semi-MultiCons* offers significant improvements. While it is more time consuming even when compared to the most efficient semi-supervised consensus clustering methods.

For the hierarchical-based *SSC*, only the *IPoptim* and *UltraTran* methods use the supervising information to learning the distance. Although the *IPoptim* and *UltraTran* can improve the clustering performance to a certain extent, they perform not so well compared with constraint-based *HAC*. It indicates that meet more constraints in the clustering process is vital. In the future work, researchers can further investigate the distance-based hierarchical *SSC* approaches, and the constraint-based approaches should also be considered simultaneously.

3.3. Density-based *SSC*

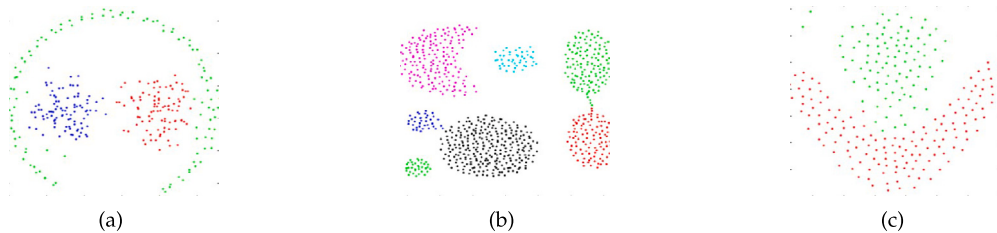
Density-based *SSC*, [72], [15], [169], [153], [116], [114], [131] contributes to enforce constraints. To improve the accuracy of clustering, must-link and cannot-link constraints among proximal samples are exploited in many approaches. These approaches are committed to being insensitive to the cluster structure assumptions and parameter setting. The specific semi-supervised density-based clustering methods are presented in this section and their detailed comparison is shown in Table 3.

Density-based clustering regards cluster as dense area of objects in data space, it can identify clusters (shown in the subgraph of Fig. 11) with different shapes [151], [152], while these methods, such as *DBSCAN*, only build optimal locally solutions. *SSDBSCAN*, [72] is the semi-supervised variation of *DBSCAN*, it aims to assign the same label to the closest unlabeled data. Once an implicit cannot-link constraint induced by the labels is detected, the extension will stop. The data objects may be left unclustered as noise as

Table 3

Comparisons of density-based SSC methods.

Method	Prior information	Time complexity	Evaluation index	Year	Advantages and disadvantages	Suitable scenario
HISSCLU [15]	class labels	$O(n^2)$	MI	2009	It can divide clusters even when there are no natural boundaries between different clusters and effective robustness to noise.	Suitable for medium to large data sets.
SSDBSCAN [72]	class labels	$O(n^2)$	RS	2009	It enables to find the structure of natural clusters although the density varies a lot between clusters and only needs one parameter while this method is order-dependent.	Suitable for datasets with large differences in density within clusters.
C-DBSCAN [116]	pairwise constraints	$O(\log^2 n)$	RI	2010	It ensures that all constraints are met, while it may generate many single sub-clusters.	Suitable to large and nosily dataset.
Active-Constrained-DBSCAN [169]	pairwise constraints	$O(n^2)$	NMI, pairwise F-measure	2012	It can actively select instance-level constraints to improve the clustering performance.	Suitable to the dataset with overlapping clusters.
An adaptive SSC approach via multiple density-based information [153]	class labels	$O(n^2)$	ACC	2017	It can determine the sets of important density-based parameters automatically, and can identify complex cluster structures in different size, shape and density adaptively.	Suitable to datasets with noise.
SSDC [114]	pairwise constraints, side information	$O(n^2)$	NMI, ARI	2018	It can find an appropriate clusters number under the guidance of prior knowledge	Suitable for dataset with arbitrary-shape cluster.
SSGC [131]	class labels	$O(n \log n)$	RI	2018	It only uses one parameter to limit nearest neighbors number.	Not suitable to data with noise and differential density.

**Fig. 11.** Clusters with different shapes. (The points with same color in each sub-graph represent the same cluster.)

enforcing constraints. Different from *SSDBSCAN*, *HISSCLU*, [15] conducts a pre-processing to guaranty that no unlabeled data is left as noise. For the data that are density-connected but have different labels, it increases the separability of objects in the pre-processing.

The *SSDBSCAN* and *HISSCLU* can efficiently discover clusters with different sizes and shapes. However, the time they take to calculate is quite expensive. To overcome this weakness, a new semi-supervised clustering, called *SSGC*, is proposed in [131]. *SSGC* employs common nearest neighbor to define the similarity among objects and considers the prior information in the clustering process. This algorithm can be effectively used to detect clusters of arbitrary shapes and different densities, while finding the number of nearest neighbors is still a problem to be solved.

C-DBSCAN, [116] is also the variation of the *DBSCAN*, and the time complexity is $O(\log^2 n)$. It is a constraint-based method which ensures that all pairwise constraints are satisfied. The process of *C-DBSCAN* is displayed in Fig. 12. The experiment in [116] shows that the performance of *C-DBSCAN* improves compared with *DBSCAN*. On all datasets, the Rand Index (RI) values of *C-DBSCAN* reaches more than 0.8. It also demonstrates its adequacy for very high performance using a small quantity of prior information (10% of the data points). Although this method is suitable to large and nosily datasets, it may retain many singleton clusters and different merge method might lead to a different result.

Another variation of *DBSCAN*, *Constrained-DBSCAN*, [169], must guarantee the constraints when adding points to a cluster. First, due to the transitivity property of the must-link constraints, the transitive closures can be obtained. Then the cannot-link constraints sets can be updated. The clustering process is repeated but still maintains the constraints until the cluster is completely discovered. In [169], *Active-Constrained-DBSCAN*, an extension of the *Constrained-DBSCAN*, was also proposed. To obtain the informative constraints set, the active selecting procedure is employed before the *Constrained-DBSCAN*. The active selecting procedure aims to obtain the

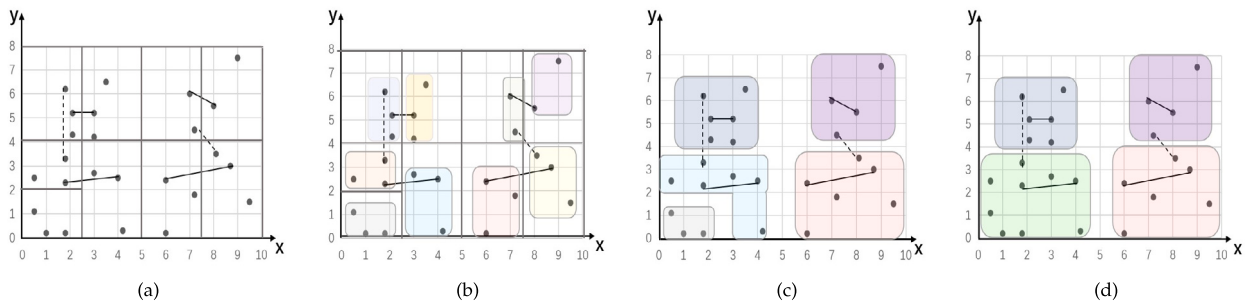


Fig. 12. An example of C-DBSCAN. (a) KD-Tree [12] is used first to obtain different partition as leaf node, each square frame represents a leaf node. (b) The local clusters (different light color areas) are created under cannot-link constraints, each data in the leaf (with cannot-link constraint) be a local cluster of its own. (c) The clusters are merged with must-link constraints in turn. (d) The final clusters can be obtained by merging clusters under cannot-link constraints.

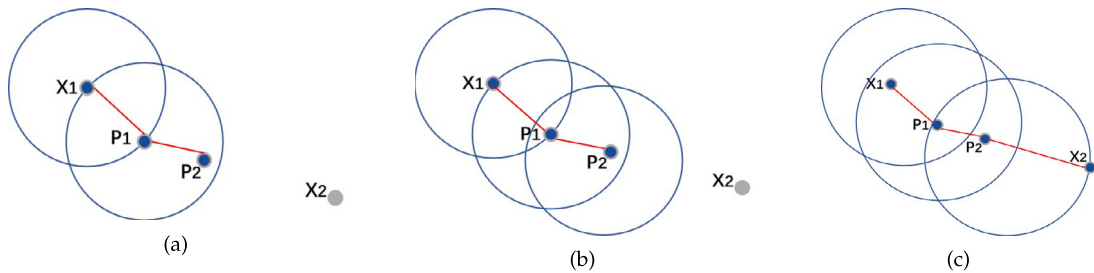


Fig. 13. An example of determining radius Eps. (a) x_1 and x_2 are two labeled samples belong to the same class k , p_1 and p_2 are unlabeled samples. A labeled data point x_1 is randomly selected as initial seed point, and its closest neighbor p_1 is assigned to the subset represented by x_1 , then let $r = \text{dist}(x_1, p_1)$ and put the data point p_2 into the subset as the distance between p_2 to any data in the subset is not bigger than r . (b) The distance among the data x_2 to any data of the subset is bigger than r , x_2 cannot be classified to the subset. (c) r is increased and the procedures are iterated until this subset contains entire data with labeled k .

constraints which contribute to determine the cluster boundaries and at a minimum one core point one core point for each cluster. The experiment in [169] demonstrated that the *Active-Constrained-DBSCAN* outperform *Constrained-DBSCAN* in all cases. Apart from that *Active-Constrained-DBSCAN* and *Constrained-DBSCAN* are suitable for the case of overlapping clusters as they inherit the advantages of *DBSCAN*.

In [153], via multiple density-based information, an adaptive semi-supervised clustering method was proposed. It determined the density-based parameters: MinPts (least data points) and Eps (radius) using both labeled and unlabeled data for each class as they have different size, shape and density. An example of determining radius Eps is shown in Fig. 13.

Experimental results in [153] show the clustering accuracy of this method on 2D-synthetic dataset up to 98.6%, while the *C-DBSCAN* performance is unstable, even with a low clustering accuracy value of 68.3%. This method can automatically determine parameters and it is quite insensitive to the noise and suitable for the complex cluster structures in different density, shape, and size. However, it is difficult to deal with high-dimensional dataset.

DenPeak, [115] is another density-based clustering, while finding an appropriate number of clusters without prior information is difficult. To solve this issue, *SSDC*, [114] produces a comparatively high amount of temporary clusters. Then it employs the pairwise constraints and side information to merge them by explicitly indicating which cluster each pair of objects belongs to.

Similar to traditional unsupervised density-based clustering, density-based SSC does not require defining the number of clusters ahead of time and can identify clusters of arbitrary shape. As the above-mentioned density-based SSC methods, they are all constraint-based, the class label or pairwise constraints can be employed to guide the clustering process. While according to our survey, there are few distance-based SSC methods. For data with high-dimensional space, the density information might become rare. Therefore, density-based semi-supervised clustering is not applicable to data with high-dimensional space. To guarantee the density information of high-dimensional data, a more accurate distance metric learning under the guidance of prior information can provide a way for researchers.

3.4. Graph-based SSC

The graph-based methods represent the samples in the form of graph. Supposing $G=(V,E)$, V stands for the set of nodes with n nodes, E denotes the edge set. In graph-based semi-supervised clustering, the label information of unlabeled nodes is learned by using the labeled nodes based on the connected edges. In view of the different number of graphs in SCC, graph-based SCC is divided into single graph and multiple graphs to solve different problems. In this section, single graph and multiple graphs semi-supervised clustering are analyzed respectively in each sub-section, and detailed analysis and comparison are provided more intuitively in the Table 4. There is a lot of literature on graph-based SCC in recent years. In order to show the table in one page, the more classical as well as the newer methods are chosen to be shown in the table, and the others are given in this paper with a brief comparative

Table 4
Comparisons of Graph-based SSC methods.

Method	Categories	Prior information	Time complexity	Evaluation index	Year	Advantages and disadvantages	Suitable scenario
NSDR-Ncut [24]	single graph	pairwise constraints	$O(n^3)$	RI, NMI	2012	It proposes a new model for generalizing instance-level constraints to space-level constraints and it can be used for clustering and segmenting images. While it has a high time complexity and space complexity.	Not suitable to large-scale data sets.
SC-PC [121]	single graph	pairwise constraints	$O(n)$	ACC	2015	It achieves high clustering accuracy, ensures scalability and filters out possible outliers and noise.	Suitable to larger datasets.
TI-APJCF/ TII-APJCFII [109]	single graph	pairwise constraints, class labels, grouping information	$O(n^3)$	ACC, NMI	2017	It is fully compatible with the three well-known categories of supervision, and can self-adapt to any semi-supervised scenario. Apart from that, it has strong robustness to the number of pairwise constraints and the parameter. However, the computing on large data sets is also very burdensome.	Suitable to medium- and small-scale semi-supervised data sets.
AHMRf-SC [37]	single graph	pairwise constraints	$O(n^2)$	CRI [67]	2018	It has high efficiency, less memory consumption and more robust to the noise data. But when there are too many constraints, it will take a long time to find the appropriate cluster centers.	Suitable to large-scale nonlinear data.
AMSS [82]	multiple graphs	class labels	$O(n^2)$	ACC, NMI	2019	It integrates various information seamlessly and performs feature extraction during the learning process, while some valuable attributes have not been found.	Not suitable to larger data sets.
NLPPC [5]	single graph	pairwise constraints, class labels	$O(n^2)$	NMI, ARI	2020	It solves the misalignment between the pre-given labels and clustering labels.	Suitable to high-dimensional datasets.
MGSC [83]	multiple graphs	class labels	$O(n^2)$	ACC, RI	2021	It proposes a constrained optimization problem that can determine cluster structures and degrees of association simultaneously in multiple graph clustering.	Suitable to larger data sets.
RSSC [173]	single graph	class labels	$O(n^3)$	NMI, ACC	2022	It can achieve robust performance of clustering even when the amount of noise is large as it identifies noise well, while the clusters number needs to be defined manually and the time complexity is relatively high.	Not suitable to larger data sets.
SSCCDS [76]	single graph	class labels	$O(n^3)$	NMI, ACC, F-measure	2022	It adds two complementary regularization terms to make full use of label information and can make clustering performance not affected by low dimensional representation, but the parameter selection deserves further study.	Suitable to low dimensional data analysis.
Adaptive and structured graph learning for SSC [84]	single graph	pairwise constraints	$O(n^3)$	ACC, NMI	2022	It fully considers the graph structure to achieve the optimal clustering performance, and it is more accurate and robust to noise.	Not suitable to larger data sets.
CSSC-SC [60]	single graph	class labels	$O(n^3)$	NMI, RI	2022	It can find the optimal number of cluster.	Not suitable to larger data sets.

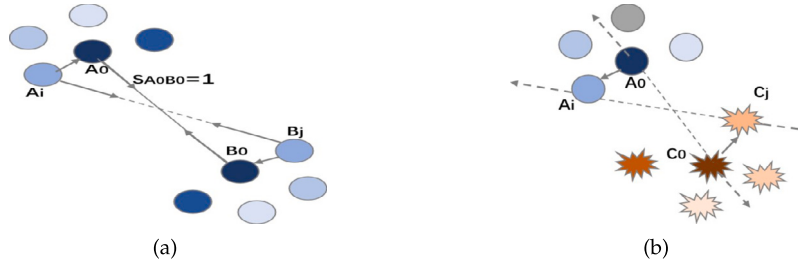


Fig. 14. The NSDR model. (a) A_0 and B_0 are relatives, though A_0 and B_0 live far away, the constraint among their force their neighbors close. $S_{A_i B_i}$ is the similarity value between A_i and B_i , it can be calculated by $\max \{S_{A_i B_i}, S_{A_i A_0} \cdot S_{A_0 B_0} \cdot S_{B_i B_0}\}$. (b) A_0 and C_0 are near strangers, the constraint among their pushes their neighbors far from each other, $S_{A_i C_i}$ can be calculated by $\min \{S_{A_i C_i}, 1 - S_{A_i A_0} \cdot S_{C_i C_0}\}$.

description. The following mentioned literature introducing the theory of spectral clustering to improve graph-based SCC, producing constrained spectral clustering. These clustering methods inherit the distinctive merits of spectral clustering such as having global optima and fitting convex and non-convex data sets.

3.4.1. SSC based on single graph

The standard framework of constrained spectral clustering for a single graph can be constructed as follows:

$$\min \sum_{i,j=1}^n w(i,j)(f_i - f_j)^2 + \lambda \sum_{i=1}^n (f_i - y_i)^2 \quad (14)$$

where $\lambda \geq 0$, $w(i,j)$ denotes the weight among the i -th node to the j -th node, f_i is the label indicator vector of the i -th node, if the label of i -th object is k , set f_i to 1, otherwise, 0, y_i stands for the given label of the i -th object. The first regularization term is derived from spectral clustering, it indicates the smoothness of f_i on the graph. And the second regularization term makes the obtained labels consistent with given labels.

Different methods can be applied to obtain the solution of eq. (14). Most methods transform it to Laplacian matrix decomposition to generate a lower dimensional embedding under the help of prior information. The Laplacian matrix is constructed by the diagonal degree matrix and the weighted adjacency matrix. In addition, some new technologies are proposed to calculate the optimal clustering results of constrained spectral clustering by some other methods. Based on the above analysis, this subsection is organized into two parts: (1) generate clustering result by decomposing a new Laplacian matrix [24], [87], [109], [5], [173], [76], [84], (2) generate clustering result by other methods [37], [121].

(1) In the methods of decomposing a Laplacian matrix, considering prior information to obtain a new Laplacian matrix is effective. The methods of decomposing a Laplacian matrix transform eq. (14) as:

$$\min_{\mathbf{F}} (\mathbf{F}^T \mathbf{L} \mathbf{F}) + \lambda \|\mathbf{F} - \mathbf{Y}\|_F^2 \quad (15)$$

in which $\mathbf{L}$, defined as $\mathbf{L} = \mathbf{D} - \mathbf{W}$, denotes the graph Laplacian matrix, $\mathbf{D}$ denotes a diagonal degree matrix, i.e. $\mathbf{D} = \text{diag}(d_1, \dots, d_N)$ and the diagonal elements d_i denotes the total weights of the i th node to other node, i.e. $d_i = \sum_{j=1}^N W(i,j)$. $\mathbf{F} \in \mathbb{R}_{N \times K}$ denotes the label indicator matrix corresponding to f_i , $\mathbf{Y} \in \mathbb{R}_{N \times K}$ is a known label matrix corresponding to y_i . By the calculation of $\mathbf{F} = \lambda(\lambda \mathbf{I} + \mathbf{L})^{-1} \mathbf{Y}$, eq. (15) can obtain a closed-form solution.

To generate a new Laplacian matrix, a method called *NSDR-Ncut*, modified the similarity matrix to space-level constraints by extending the instance-level constraints, was presented by [24]. An example for the proposed *NSDR* model is displayed in Fig. 14. Based on experiments in [24], *NSDR-Ncut* is more robust on diverse datasets. While in some dataset, *Cop-Kmeans* is comparable with *NSDR-Ncut*, such as on *SOY BEAN* dataset, *Cop-Kmeans* performs best in Rand Index, however, *NSDR-Ncut* performs best in NMI. In the consuming of time and space, the generalized eigenvalue decomposition is time-consuming and requires a great deal of space to storage the affinity matrix.

Similarly, in order to obtain a new similarity matrix, two methods (*TI-APJCF* and *TII-APJCF*) with affinity and penalty jointly constrained spectral considered the penalty of constraints in the similarity matrix and obtained the diagonal matrix according to the similarity matrix, they were proposed in [109]. *TI-APJCF* and *TII-APJCF* are fully compatible with three well-known types of supervising information. The feasible conversions among the three types of supervision are shown in Fig. 15. In view of running time, *TI-APJCF* and *TII-APJCF* were averagely in moderate levels compared with the comparison methods in [109]. This is mainly as these methods compute the eigenvalue decomposition issues by employing the *MATLAB* built-in function-eig(). *RSSC*, [173] uses a new similarity matrix to address the problem of noise interference. In the process of generating a new similarity matrix, the label information is used to combine with the normalized graph Laplacian matrix obtained from the original dataset. In literature [84], a new semi-supervised clustering is proposed by learning two similarity matrices in feature space and label space. Instead of creating a new similarity measure directly, *SSCCDS*, [76] imposes similarity and dissimilarity two regularization items. While reducing the time complexity to adapt to large data sets is still a problem to be solved in the future.

Although the methods above mentioned achieve better performance, they not take into account the mismatch among the preset class tags and clustering tags (as shown in Fig. 16).

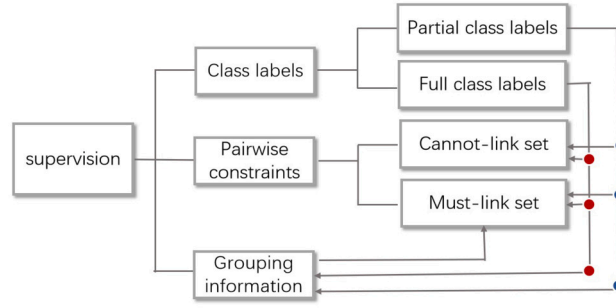


Fig. 15. Three categories of supervising information and workable transformations between them. (Partial class labels and full class labels can be easily transformed to the must-link constraints or cannot-link constraints and the grouping information, the grouping information can be converted into the must-link constraints as two members in each group belongs to the same cluster.)

Classes Objects	Class 1	Class 2	Class 3	Class 4
X_1	1	0	0	0
X_2	0	1	0	0
X_3	0	1	0	0
X_4	0	0	1	0
X_5	0	0	0	1

(a)

Class 1 $\Rightarrow$ Cluster 4
 Class 2 $\Rightarrow$ Cluster 3
 Class 3 $\Rightarrow$ Cluster 1
 Class 4 $\Rightarrow$ Cluster 2

(b)

Clusters Objects	Cluster 1	Cluster 2	Cluster 3	Cluster 4
X_1	0	0	0	1
X_2	0	0	1	0
X_3	0	0	1	0
X_4	1	0	0	0
X_5	0	1	0	0

(c)

Fig. 16. Mismatch among the pre-set class tags and the cluster tags: (a) is the pre-given matrix Y , X_1 is given the label of class1; (c) is the obtained membership matrix F , X_1 belongs to cluster4; (b) is the misalignment of the relation between classes and clusters.

NLPPC, [5] used the divergence between the pairwise relations and clustering results as a penalty factor to construct a new optimization model to solve the misalignment problem, it can effectively accommodate high-dimensional datasets.

The experiments result in [5] show that *NLPPC* outperforms the other methods using various kinks of prior information. With help from positive labels, *NLPPC* is more accurate than *LNP*, [99] (defined the edge weights by using these linear neighborhoods to replace the Gaussian kernel function), which indicates *NLPPC* can improve the clustering effectiveness by solving the mismatch problem among class tags and cluster tags. Compared with *PNLP*, [176], a positive and negative label propagation method, *NLPPC* is obviously superior on most of the datasets with positive and negative label, on the other datasets such as *Isolet*, the performance closely approximates *PNLP*. It can also show that *NLPPC* is insensitive to positive and negative label. When considering the prior information of pairwise constraints. *PCLP*, [176], achieving the propagation of the pairwise constraints in both the row and column directions, has slightly high accuracy than *NLPPC* on the *Heart* dataset with 15% pairwise constraints, while *NLPPC* has more accurate on most datasets.

(2) Constrained spectral clustering can also obtain the result by other methods. Referring to the mathematical connection between *HMRF* semi-supervised clustering and approximate weighted kernel k-means, the optimal results of *HMRF* spectral clustering in [37] is calculated by using the approximate weighted kernel k-means. This method has better time performance and can be directly applied in large scale datasets.

While the method obtained by decomposing a Laplacian matrix needs to construct the affinity matrix, the Laplacian matrix calculation requires $O(n^2)$ time and eigenvalues and eigenvectors computation need $O(n^3)$ time.

[121] proposed the *SC-PC* method to inject pairwise constraints to a small affinity submatrix, then performed constraints propagation in the affinity submatrix to increase the clustering performance. The obtained highly reduced matrix performs Singular Value Decomposition (*SVD*) to generate clusters. *SC-PC* achieves better *ACC* and *NMI* than other methods compared in [121] by the dynamic fitness and propagation the constraints to the local neighborhood. In terms of the time-consuming, the calculation costs of *SC-PC* are relatively low, especially for large-scale data set of *MNIST*. According to the experimental results in [121], *SC-PC* is at minimum 40 folds faster than *NCut*, [122].

3.4.2. SSC based on multiple graphs

Multiple graphs *SSC*, [82], [83] divides data point into different clusters depending on the knowledge provided by different graphs, so that the clusters in one graph are separated well and the clusters in various graphs are more consistent. As it integrates information from multiple graphs compared to single graph clustering, the clustering performance can be enhanced. These methods obtain more and more attention as its dominant advantages in clustering precision.

AMSS, [82], a seamless integration semi-supervised graph clustering technique for multi-view data, consists of the following two processes: (1) multi-view fusion, this step fuses the information of node attributes and various prior information. (2) auto-weighted learning for node similarity, it contains the step of extracting discriminative features. *AMSS* extracts discriminative features to improve its clustering performance, so that, it is useful when the graph contains erroneous information. The experiments in [82]

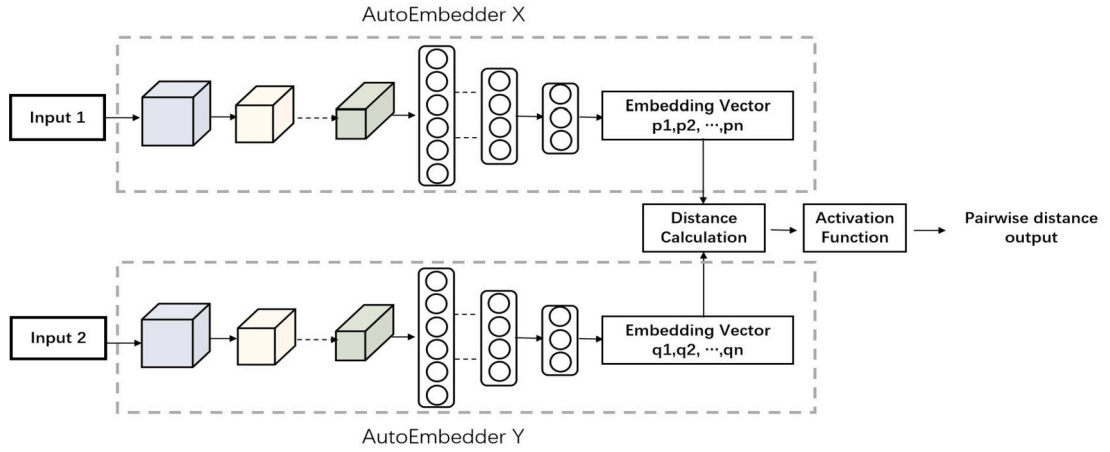


Fig. 17. The SNN training architecture.

proved that AMSS had superior clustering performance with increasing noise. In practice, the association between some graphs to the other graphs may be strongly or weakly.

To take the association degree among different graphs into account, MGSC, [83] was proposed. MGSC is a constrained optimization problem that uses cosine correlation and orthogonality respectively to determine the association degree and cluster structures. It considers multiple graphs SSC approach by minimizing eq. (16):

$$\begin{cases} \min J_{MGSC} = - \sum_{m=1}^M \text{tr}(F_m^T A_m F_m) - \sum_{m=1}^M \sum_{m'=1, m' \neq m}^M \beta |\text{tr}(F_m F_{m'}^T)| \text{tr}(F_m F_m^T) + \lambda \sum_{m=1}^M \|F_m - Y_m\|_F^2 \\ \text{s.t. } F_m^T F_m = I_k (1 \leq m \leq M) \end{cases} \quad (16)$$

The first term and the second term denote the loss function of multiple graphs clustering, in which F_m is the label indicator matrix of the m -th graph. In the m -th graph, if the i -th object's label is k , then set $F_m(i, k) = 1$, otherwise, $F_m(i, k) = 0$. A is the block adjacency matrix which includes the information in each graph and the association between different graphs, β is positive and it controls the balance among the cluster separability in each graph and cluster consistency among different graphs. The third is a cost term of inconsistency of the obtained labels and the given labels, $\lambda \geq 0$ and λ stands for a regularization parameter.

Experiments in [83] demonstrated that the compared multiple graphs methods except AMGL, [102] have a better result than conventional semi-supervised single graph clustering. It indicates that multiple graphs consist of more comprehensive objects information and the information integrated by multiple graphs can improve the clustering result. Compared with the other methods, MGSC obtains a higher RI/ACC value under different rates of given label.

In the graph-based SSC, the constraints-based methods are performed by adding the penalty term, and the distance learning-based methods are generally realized by modifying the Laplacian matrix. The simultaneous use of both methods can be pursued as future work. In addition, most of the graph-based SSC methods have high time complexity and are not applicable to larger data sets. Therefore, future research work should pay more attention on reducing the time complexity.

3.5. Neural network-based SSC

The neural network-based SSC introduced in this section designs to guide the clustering by incorporating prior information into neural networks. Deep neural networks (DNN) are widely applied in clustering, the methods of using DNN on clustering are called as deep clustering. [126] divided the deep clustering into two phases, the first phase is feature transformation using DNN and then the second phase is clustering. While some researchers are dedicated in making use of deep clustering method to learn feature transformation and clustering jointly. For this reason, the neural network-based SSC enhances the overall performance of deep clustering by (1) learning feature transformation and clustering separately (2) learning feature transformation and clustering jointly.

(1) To achieve the goal of improving the performance of deep clustering, some improved methods are given by dividing the deep clustering into two phases [104], [124], [178], [165]. [104] uses pairs of data with pairwise constraints as input of Siamese Neural Network (SNN) (shown in Fig. 17) to train AutoEmbedder (The SNN model consists of two AutoEmbedder with the same weight in parallel). Then, the well trained AutoEmbedder can be used to generate meaningful embeddings. Finally, the clustering is performed on the obtained embeddings. For any pair of the input data x_i and x_j , if there are cannot-link constraint between them, the pairwise constraint is evaluated as be α (a distance hyperparameter), otherwise, the value of distance is estimated to 0.

This method is applicable to high-dimensional dataset. The experiences in [104] compare the ACC, NMI, and ARI scores of several methods on different image datasets and one textual dataset. The experimental results show that unsupervised methods do not perform well on CIFAR-10 and Fashion-MNIST complex datasets. Although methods based on GAN with semi-supervised [124] and

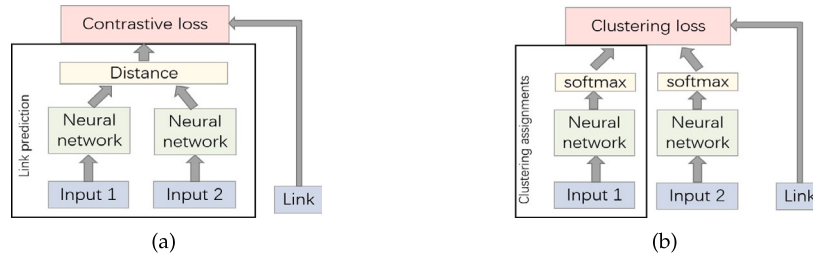


Fig. 18. An illustration of the proposed $ss-S^3C^2$ model. (a) LabNet: labeling network, (b) CluNet: clustering network.

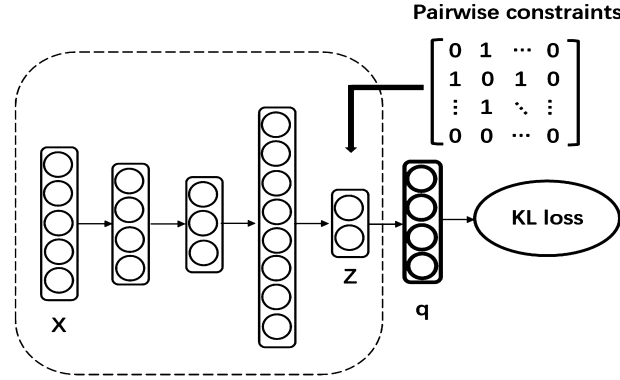


Fig. 19. The framework of SDEC.

unsupervised technology [99] perform better, they may be affected by the unstable learning of generators and discriminators [124], while the *AutoEmbedder* does not.

Discriminative clustering methods, using classification approach to solve clustering problems, have been shown usually more effective in using label or pairwise constraints information. For this purpose, Marek mieja et al. used two Siamese neural networks to generate a classification-based approach, called $ss-S^3C^2$, [178]. *LabNet* is the first network used to classify whether must-link or cannot-link exists in two samples. As Fig. 18(a) shows, the training phase only uses pairwise constraints. If $(x, y) \in M/C$, the *LabNet* uses the contrastive loss to make the return representation $h(x)$ close/distant to $h(y)$. Once the *LabNet* is trained, it is utilized to generate pairwise constraints between all pairs of samples. The second network, *CluNet*, indicates in Fig. 18(b), predicts final clusters assignments on labeled pairs. The *CluNet* is instantiated as a Siamese pair with the same DNNs, softmax output layers yield class posterior probabilities. In training phase, only pairwise constraints are used. During the testing phase, producing cluster assignments only needs one of the networks.

Based on the performance in [178], in the case of using small number (100 and 200 links) of constraints, *d-graph*, [123] obtains the largest value of *NMI*. When the number of constraints get higher, *d-graph* outperforms $ss-S^3C^2$ and *IDECA*, [165], owing to that *d-graph* only uses distances to generate auxiliary labeled pairs. The performance of *IDECA* is unstable, the outcomes are not consistently growing with the amount of constraints increase on the datasets of *MNIST* and *Letters*. Although S^3C^2 has the lower *NMI*, it is being partly offset by semi-supervised variant ($ss-S^3C^2$). $ss-S^3C^2$ generates a remarkably nice outcomes on the *Letters* data set, which consists of 26 classes.

(2) Learning feature transformation and clustering jointly is another way to strength the deep clustering [142], [113]. There are some previous reports using *DEC*, [142]. *DEC* learnt a mapping utilizing a joint optimization approach to transfer the feature space to a lower one. Then in this transformed space an effective objective was optimized. Despite the success of *DEC* in clustering, it cannot further improve the clustering performance as it fails to utilize such prior knowledge in clustering process. *SDEC*, [113], a semi-supervised variant of the *DEC*, was demonstrated to have better accuracy. Fig. 19 shows the framework of *SDEC*, it first initializes the DNN structure by using the encoder layers of a pre-trained stacked auto-encoder [129], then *SDEC* directly adds the pairwise constraints to the embedding layer Z to learn the new feature representation. Finally, the KL divergence loss is calculated by the obtained soft assignment q of each data point.

In the experiments of [113], *SDEC* performs better than the unsupervised deep embedded clustering method of *DEC* and *IDECA*, [165]. Specifically, the *ACC* and *ARI* value of *SDEC* significantly improves on the dataset of *20NG* compared with *DEC*. This indicates that only use a small number of prior knowledges can enhance the clustering performance by a large margin.

The neural network-based SSC uses the label data or pairwise constraints to constraint the distance learning or clustering process. Such as in [104], [178], to obtain the distance, the siamese neural network is used and the pairwise constraints are employed to limit the output of pairwise distance. Although adding supervising information in deep clustering has achieved good results, the current research results in this area are still relatively small, and we can try to do further research.

3.6. Nonnegative matrix factorization-based SSC

The data dimension is extremely high in real life. The existence of noise and redundancy may be severely degraded the performance of traditional clustering methods [19]. Apart from that, the attributes of high dimensional datasets are too many, it may cause the overfitting problem. For adapting *SSC* to the demand of high dimensional data, matrix factorization is employed. Among existing matrix factorization techniques, *NMF*, [71] receives an increasing attention. Based on matrix factorization, the research is carried out as follows.

To enhance the performance of *SSC* in high-dimensional data, *NMF* is used first. One semi-supervised version of *NMF*, called *CNMF*, was proposed in [81]. It introduces a matrix C with N_L row and S column to integrate label information, in which N_L and S denote the labeled data and the exist classes' number. If x_i is labeled with class j , $C_{i,j} = 1$, otherwise, $C_{i,j} = 0$. The label constraint matrix A is given based on the matrix C :

$$A = \begin{bmatrix} C & 0 \\ 0 & I_{N-N_L} \end{bmatrix} \quad (17)$$

where I_{N-N_L} is an identity matrix of $(N - N_L) \times (N - N_L)$. One popular cost function of *CNMF* is shown in eq. (18).

$$\begin{cases} \min J_{CNMF} = \|X - U(AZ)^T\|^2 \\ \text{s.t. } U \geq 0, AZ \geq 0 \end{cases} \quad (18)$$

CNMF uses the multiplication of non-negative matrices U and V to approximate matrix X , in which U is considered as a group of basis vectors, V shows the coordinates of each sample of U , which is expressed by multiplying the matrix A and Z .

While in the new representation of *CNMF*, samples have same label will disappear to a single data, it maybe undesirable in some real applications. Based on the analysis, the improved *NMF*-based *SSC* are classified in the basis of it is performed on single view or multiple views. The improved *NMF*-based *SSC* method based on *CNMF* are shown in Table 5 (in exception to *CNMF*).

(1) For single view, semi-supervised *NMF* methods is divided into explicitly or implicitly add prior information.

Explicitly [4], [75]: The first one is to explicitly add prior information by considering them as hard constraint into the objective function. *DNMF*, [4] utilized a fraction of label information to generate a discriminative constraint. The label information is used to obtain a matrix Q , if the data point j is labeled and it belongs to the i th class, set $Q_{i,j} = 1$, otherwise $Q_{i,j} = 0$. Assuming that $N = 8$, the number of labeled data $N_L = 3$, the labeled data with the sample categories of $L_{x_1} = 3$, $L_{x_2} = 1$, $L_{x_3} = 2$, Q is a matrix with 8 rows and 3 columns. In matrix Q , the element of the first row and the third column, the second row and the first column, and the third row and the second column are 1.

According to the matrix Q , the objective function of *DNMF* is proposed (as shown in eq. (19)). *DNMF* uses this auxiliary matrix B to transform and scale the vectors in the new representation in a linear way. The intrinsic geometric structure of the samples is not considered in *DNMF*, which can be solved by using the local manifold regularization. Motivated by *DNMF* and manifold regularization, semi-supervised graph-based *GDNMF* method [75] was proposed. The objective function of *GDNMF* is formulated as eq. (20):

$$\begin{cases} \min J_{DNMF} = \|X - UV^T\|^2 + \alpha \|Q - BV_L^T\|^2 \\ \text{s.t. } U \geq 0, V \geq 0 \end{cases} \quad (19)$$

$$\begin{cases} \min J_{GDNMF} = \|X - UV^T\|_F^2 + \alpha \|Q - BV_L^T\|^2 + \beta \text{Tr}(V^T L V) \\ \text{s.t. } U \geq 0, V \geq 0 \end{cases} \quad (20)$$

in which $V_l = [v_1, \dots, v_{N_L}, 0, \dots, 0]$, and $V_l \in R^{N \times k}$, the auxiliary matrix $B \in R^{S \times k}$. For sake of the best fit of the matrix Q , α and β are tradeoff parameters for balancing the regularization terms, L is graph Laplacian matrix.

Based on the experimental in [75], the *ACC* and *NMI* value of *GDNMF* respectively increase by 51.77% and 58.70% compared with *GNMF*, [17] on the *PIE* face database. This is because *GDNMF* is a semi-supervised learning method while *GNMF* is a completely unsupervised learning method. As the usage of the graph regularization in *GDNMF*, the *ACC* and *NMI* value of *GDNMF* can respectively achieve 2.60% and 2.45% better than *DNMF* on *TDT2* corpus.

In *CNMF*, the points that have same label are represented as one point. While the data points in *DNMF*, which belong to the same class, should be very close or align on the same axis in the new representation. They did not merge into a single point. However, *CNMF* and *DNMF* did not reveal the intrinsic geometric structure of the data, *GDNMF* solves this problem by introducing manifold regularization. In future, the performance of *GDNMF* should be further improved, such that it can be applied to many fields such as face recognition, image clustering, and *DNA* gene expressions.

Implicitly [74], [108]: The second one is to implicitly incorporate pairwise constraints into the weight matrix. *PGCNMF*, [74] adds prior knowledge to the data local construction via graph Laplacian and the convex-*NMF* framework. As the usage of convex-*NMF* framework, it can also handle mixed-sign matrix, which extends the application of *NMF*. The objective function of *PGCNMF* is performed as eq. (21).

$$\begin{cases} \min J_{PGCNMF} = \|X - XUV^T\|_F^2 + \lambda \text{Tr}(V^T L V) + \alpha \text{Tr}(V^T M V A + V^T C V) \\ \text{s.t. } U \geq 0, V \geq 0 \end{cases} \quad (21)$$

Table 5
Comparisons of nonnegative matrix factorization-based SSC.

Method	Prior information	The way to add Prior information	Time complexity	Evaluation index	Year	Advantages and disadvantages	Suitable scenario	Input data type
DNMF [4]	class labels	explicitly	$O(nmK)$	ACC, NMI	2016	Its convergence speed is equivalent to that of <i>NMF</i> and the number of data points is not decreased. But it is not extending to the divergence formulation and adding the locality preserving property to the DNMF.	Suitable to noisy data.	feature matrix, label matrix
GDNMF [75]	class labels	explicitly	$O(nmK \cdot T + n^2m)$	ACC, NMI	2017	Its takes the available label information and local manifold regularization into account.	Suitable to large and high-dimensional datasets.	feature matrix, label matrix, Laplace matrix obtained from weight matrix and degree matrix
PGCNMF [74]	class labels	implicitly	$O(nmK \cdot T + n^2m)$	ACC, NMI	2017	It can deal with mixed-sign data, and the obtained basis and encoding vectors have better representation power. While this method is sensitive to parameters.	Suitable to high-dimensional datasets and datasets with mixed-sign data.	feature matrix, label matrix, Laplace matrix obtained from weight matrix and degree matrix, in which class label information is added to weight matrix
CMVNMf [177]	pairwise constraints	explicitly	$O((2v+1)kn^2 + 2nmk)$	NMI	2018	It solves the problem of multi-view clustering for unmapped data. However, the computational time is increased.	Suitable to datasets with high-dimensional data.	feature matrix, constraints matrix
MVCNMf [18]	class labels	implicitly	$(n^v tm^v nk)$	ACC, NMI, F-measure	2019	It can effectively guarantee that the consistency of the label information in the new data representation and the original data. While it is sensitive to the parameter.	Suitable to datasets with high-dimensional data.	feature matrix, label matrix
MVOCNMf [50]	class labels	implicitly	$O(tk(2m^v n + n^2))$	ACC, NMI, F-measure	2020	It obtains the desirable representations for each view. While the computational time is increased.	Suitable to datasets with high-dimensional data.	feature matrix, label constraint matrix, constraint matrix
GPSNMf [79]	class labels	implicitly	$O(tm^v nk)$	Precision, Recall, ARI, F-measure, Purity, ACC, NMI	2020	It incorporates the label information and preserves the intrinsic geometric structure of the data. However, it is sensitive to the parameter.	Suitable to datasets with high-dimensional data.	feature matrix, Laplace matrix
CSNMf [108]	pointwise, pairwise constraints	implicitly	$O(nmK \cdot t + n^2m + n^2p)$	ACC, NMI, Purity	2021	It suppresses the influence of non-Gaussian noise or outliers contaminated in real world data, and uses the pointwise and pairwise constraints to obtain the discriminative data representation. While it is difficult to achieve the global minima.	Suitable to datasets with high-dimensional data and non-Gaussian noisy data.	feature matrix, label matrix, Laplace matrix
semiEMC [49]	pairwise constraints	implicitly	$O(Ttvmnk + pmn^2 + n^3)$	ACC, NMI, Purity	2021	It can achieve exclusive clustering interpretation, preserve the correct geometric structure and fully use the prior information. While the disadvantage is the weights among different views were treated equally.	Suitable to datasets with high-dimensional data.	feature matrix, label matrix
DHPS-NMF [164]	class labels	explicitly	$O(tm^v nk)$	ACC, NMI, ARI, F-measure, Precision, Recall	2022	It preserves the higher-order relationships in the data, and considers the consistent and complementary information between different views.	Suitable to datasets with high-dimensional data but takes long time.	feature matrix, label matrix

Notations:

m, K : the dimension of the feature vectors, the new reduced dimension.

p : the number of nearest neighbors on the graph.

T : the iteration of the outer loop in [49].

v, m^v : the number of views and the dimension of each sample in v th view.

in which λ and α are two positive parameters, they respectively indicate the importance of the graph structure and the violation of the pairwise constraints, L is the graph Laplacian matrix, it is calculated from the degree matrix and the weight matrix, and the label information is added to the weight matrix. M and C represent label indicator matrix respectively, A denotes a $k \times k$ auxiliary matrix, its diagonal elements are 0 and others are 1.

The experiments in [74] show that *PGCNMF* can achieve good performance, while it is parameters λ and α sensitive. Such as on the *Yale* date set, α from 0 to 20 with λ is fixed, *PGCNMF* can achieve good and stable performance consistently, while it drops if α is bigger than 20. When λ varies from 0 to 10 with α is fixed, *PGCNMF* can achieve relatively stable performance, while the *ACC* and the *NMI* drops respectively if λ is greater than 10. However, the selection of suitable parameters values in a theoretical way remains an issue.

CSNMF, [108] not only utilized correntropy as the similarity measure to replace the squared Euclidean distance (*SED*) (as shown in the first term of eq. (22)) in semi-supervised *NMF*, but also fully used the limited labeled data points to generate more supervising information (as shown in the second term of eq. (22)). This method can address the heavy-tailed non-Gaussian noises or outliers issues.

$$\text{Max} J_{\text{CSNMF}} = \sum_{i=1}^M g\left(\sqrt{\sum_{j=1}^N (X_{ij} - (UZ^T A^T)_{ij})^2}\right) - \beta \text{Tr}(Z^T A^T \tilde{L} A Z) \quad (22)$$

in which $g(a-b) = e^{-\frac{2\sigma^2}{(a-b)^2}}$, A is a label constraint matrix, its definition is the same as that in *CNMF*, Z is an auxiliary matrix, β is the graph regularization parameter with $\beta \geq 0$, $\tilde{L}$ indicates the graph Laplacian matrix with $\tilde{L} = \tilde{D} - \tilde{S}$, $\tilde{S}$ indicates the new edge weight matrix learned from *CPA*, [134], [86], $\tilde{D}$ is a diagonal matrix.

As the experiments in [108], *CSNMF* achieves best performance compared with other methods. Especially, in view of *NMI*, *ACC*, and *Purity*, *CSNMF* can improve 4.57%, 8.06%, and 0.34% respectively compared with *CPSNMF*, [36] on *MSRA25* data set. It is owing to *CPSNMF* is a semi-supervised *NMF* approaches based on conventional *SED*, and *CSNMF* simultaneously utilizes two different types of prior information.

(2) For multiple views [177], [18], [50], [49], [79], [164], *SSC* aims to obtain an accurate clustering result by exploring the compatible and complementary information embedded in the heterogeneous multi-view samples. Considering that judging a must-link or cannot-link constraint is considerably simpler than recognizing a mapping. *CMVNMF*, [177] is proposed on unmapped data, the objective function of *CMVNMF* is shown in eq. (23).

$$\begin{cases} \min J_{\text{CMVNMF}} = \sum_{a=1}^V (\|X^a - U^a (V^a)^T\|_F^2 + \frac{1}{2} \mu_{a,a} \Delta_{a,a}) + \sum_{a,b=1, a>b}^V \mu_{a,b} \Delta_{a,b} \\ \text{s.t. } U^a \geq 0, \|U_j^a\|_1, j \in \{1, \dots, l\}, V^a \geq 0, \sum_{a,b=1, a \geq b} e^{-\mu_{a,b}} = 1 \end{cases} \quad (23)$$

$\Delta_{a,b}$ denotes the inconsistency among view a and view b , through setting $a = b$, it can also indicate the bias of two samples from one view by utilizing intra-view constraints, V_a represents an indicator matrix of the a th view, $\mu_{a,b}$ controls the importance of each view pairs, U_j^a is the j th column of U^a .

Using intra-view and inter-view constraints simultaneously, the *NMI* value increases as constraints number increases in the experiments of [177]. With equal amount of constraints, the *NMI* value grows as the inter-view constraints increases, and achieves the best scores only using constraint between views. The main reason is inter-view constraints can provide complementary information among different views. Although the performance is increased, the computational time is also increased.

In addition to the method mentioned above, most existing approaches employ graph regularized *NMF* to separately obtain the new representation of each view. However, it is no guarantee that the obtained representation has the same label with the original data. In [18], a new method based on *CNMF* with sparseness constraint (*MVCNMF*) is proposed. For each view, this method introduced the auxiliary matrix. Then, the common consensus label constraint matrix was constructed, it shared by all views to guarantee that the label information was fused into the new representation of each view. After that, to ensure different view information complementary with each other as much as possible, the co-regularization was imposed in the auxiliary matrices. The objective function of *MVCNMF* is shown in eq. (24).

$$\begin{cases} \min J_{\text{MVCNMF}} = \sum_{v=1}^{n_v} \theta_v \|X^V - U^V (Z^V)^T A^T\|_F^2 + \sum_{v=1}^{n_v} \sum_{S=1}^{n_v} \theta_{vs} \|Z^V - Z^S\|_F^2 + \lambda \sum_{v=1}^{n_v} \|Z^v\|_{2,1} \\ \text{s.t. } U \geq 0, V \geq 0 \end{cases} \quad (24)$$

in which A is the label constraint matrix and Z^v is the auxiliary matrix from different views, the first term takes into account every view, θ_v denotes the v th view's weight, and the second term denotes the relationship of similarity among different views, θ_{vs} denotes the weight of the similarity among Z^v and Z^s , last term imposes $\ell_{2,1}$ -norm on the auxiliary matrix, and λ controls the sparseness constraint. The integration of the sparseness constraint and the co-regularization avoids the views with different quality influencing with each other, and improves each view's clustering performance.

While the experiments in [18] show the parameter affects the performance of *MVCNMF*, such as on 3Sources dataset, when $\theta_v \leq 0.01$, it keeps steady increase with θ_{vs} increase, and when $0.01 < \theta_v < 10$, it decreases with the increase of θ_{vs} .

Many semi-supervised clustering methods based on graph regularization only take pairwise relationships among samples into account. *DHPS-NMF*, [164] constructs samples and feature hypergraphs to learn the higher-order relationships existing in the sample manifold and feature manifold of multi-view data. In order to effectively use the label information to improve the discrimination of the obtained low-dimensional features, semi-supervised regression terms are incorporated into the model when capturing the complex manifold structure of the data.

Apart from fusing the label information into the new representation of each view, [50] obtained the desirable representations for each view, it proposed a *MVOCNMF* method represented by an objective function (as shown in eq. (25)).

$$\begin{cases} \min J_{MVOCNMF} = \sum_{v=1}^{n_v} \theta_v \|X^V - U^V (Z^V)^T A^T\|_F^2 + \lambda \sum_{v=1}^{n_v} \|P \circ Z^V (Z^V)^T - I\|_F^2 + \sum_{v=1}^{n_v} \sum_{s=1}^{n_v} \frac{1}{2} \theta_{vs} \|Z^V - Z^S\|_F^2 \\ \text{s.t. } U^V, Z^V \geq 0 \end{cases} \quad (25)$$

where the first term learns the samples' low-dimensional representations of each view, and the second term obtains the desirable representations, if $i=j$ or $1 \leq i, j \leq k$, p_{ij} in the constraint matrix P is 1, otherwise, 0, operator $\circ$ is the elementwise multiplication operator, the last term integrates the complementary information in various views. The same as *MVCNMF*, *MVOCNMF* is also easily affected by the parameters.

As shown in the experiments of [50], the *NMI*, *F-measure* and *ACC* values changed and when $\lambda = 0.1$ they reached their maximum. This observation represents that the performance can be enhanced by orthonormality constraints and *MVOCNMF* obtains the best solution when λ value is 0.1. Along with λ value greater than 0.1, all the results drop significantly. One likely explanation may be that the excessively tight orthonormality constraint reconstructs the objective data and destroys its clustering structure.

Few approaches consider exclusivity, which results in indistinct clusters. To overcome these limitations, a subspace learning approach based on *NMF* (*semiEMC*) was proposed in [49] by integrating multiple *NMF* frameworks. The double-orthogonal constraints on the exclusivity of the clusters are enforced. Particularly, the limited prior information is enriched and encoded though integrating the propagated semi-supervised manifold regularization.

Experiments presented by [49] demonstrate *semiEMC* achieves the optimal in the *NMI*, *ACC*, *Purity* value compared with five methods on seven different datasets. It is because of the following two reasons. (1) To enable the formation of diverse clusters more easily, the double-orthogonal constraints keep the different categories of data from a subspace excluded as strongly as possible. (2) To the manifold regularization reflects the real data correlation, the finite prior information are utilized in the strategy of propagation.

Based on the analysis mentioned above, the Nonnegative matrix factorization-based *SSC* can add the constraints information in the objective function of *NMF* (Nonnegative matrix factorization). They can modify the objective function by employing a penalty constraints term, or obtain a new weight matrix. Nonnegative matrix factorization-based *SSC* also can be used to handle real-world datasets with high-dimensional data and is insensitive to noise. However, the methods to add prior information in the process of calculating the distance of samples are relatively small.

3.7. Random subspace technique-based *SSC*

High dimensional datasets will lead to the over-fitting problem as they have too many attributes compared to the number of samples. Random subspace technique [52] can effectively alleviate this problem (as shown in Table 6). By introducing random subspace technique to clustering ensemble, the combination of clustering results of homogeneous or heterogeneous clustering methods makes the clustering performance greatly improved. There are two basic methods to incorporate semi-supervising information to clustering ensemble with random subspace technique. (1) Constraint-first methods introduce supervising information into cluster ensemble generation step. (2) Cluster-first methods perform ensemble generation without any supervision, while integrate limited supervision in the consensus function.

For constraint-first methods, it is rarely used by researchers as there are several disadvantages. For example, constraint-first methods frequently inefficient and inflexible to carry out semi-supervised clustering multiple times on a large dataset and generate a great number of *SSC* specially with changes in supervision.

For cluster-first methods, traditional cluster ensemble approaches have several limitations in high dimensional datasets: difficult to achieve good performance, ignorance of the different contributions among different ensemble members, the contribution of pairwise constraints to the final results is uncertain.

To perform clustering on data with high dimensional, [163] presented a *SSC* ensemble method (*RSSCE*), it combines the random subspace, constraint propagation (E^2CP) [87] and normalized cut technique within a framework of cluster ensemble. While the final result was achieved by not all members of the ensemble, most of the cluster ensemble approaches only considered use a similarity score or feature selection technique to eliminate extra members of the ensemble, and few of them study how to apply an optimization method to discover a suitable subset. Then, an incremental *SSC* ensemble framework (*ISSCE*) though adding a selection process of incremental ensemble member to remove the redundant ensemble members was designed. The framework of *ISSCE* is shown in Fig. 20(a), it first generates a set of random subspaces using random subspace technique, then E^2CP is adopted to generate the semi-supervised clustering models. After this step, *RSSCE* generates a consensus matrix to obtain the final results by summarizing the clustering solutions of E^2CP . While *ISSCE* generates the new ensemble set by the process of incremental ensemble member selection.

Experiments in [163] shows *ISSCE* outperforms other single semi-supervised clustering methods (such as E^2CP) on 17 out of 18 datasets. As most selected datasets in [163] are high dimensional datasets, the experiments results can demonstrate that *ISSCE* is able to alleviate the effect of high dimensional. The possible reasons are (1) the high dimensional data of the original space can be reduced

Table 6

Comparisons of random subspace technique-based SSC.

Method	Prior information	Time complexity	Evaluation index	Year	Advantages and disadvantages
RSSCE [163]	pairwise constraints	$O(Bpn^2 + kn^3)$	NMI, ARI	2016	It combines the random subspace, constraint propagation and normalized cut technique within a framework of cluster ensemble. While it take the ensemble members without positive contributions into account.
ISSCE [163]	pairwise constraints	$O(Bpn^2 + kn^3 + Bkn^2 + B\log B + B'nm^2)$	NMI, ARI	2016	It adds a selection process of incremental ensemble member to remove the redundant ensemble members, and it perform well on real-world datasets, especially on the high-dimensional datasets. But it is sensitive to parameter.
CESCP [162]	pairwise constraints	$O(Bpn^2 \cdot c + m^3 + cm^2 + kn^3)$	NMI	2018	It improves pairwise constraint selection and constraint projection. But it pays little attention to the clustering ensemble procedure.
DCECP [162]	pairwise constraints	$O(Bpn^2 \cdot c + m^3 + cm^2 + kn^3 + GBkn^3)$	NMI	2018	It adopts an adaptive ensemble member weighting process (AEMW) to generate the weight of the ensemble member. While it is also sensitive to parameter.
WECR k-means [68]	pairwise constraints	$O(sn^2(k + m))$	NMI, ARI	2019	It is much more robust to noisy constraints, and it improves the computation efficiency and robustness of semi-supervised cluster ensemble.

Notations:

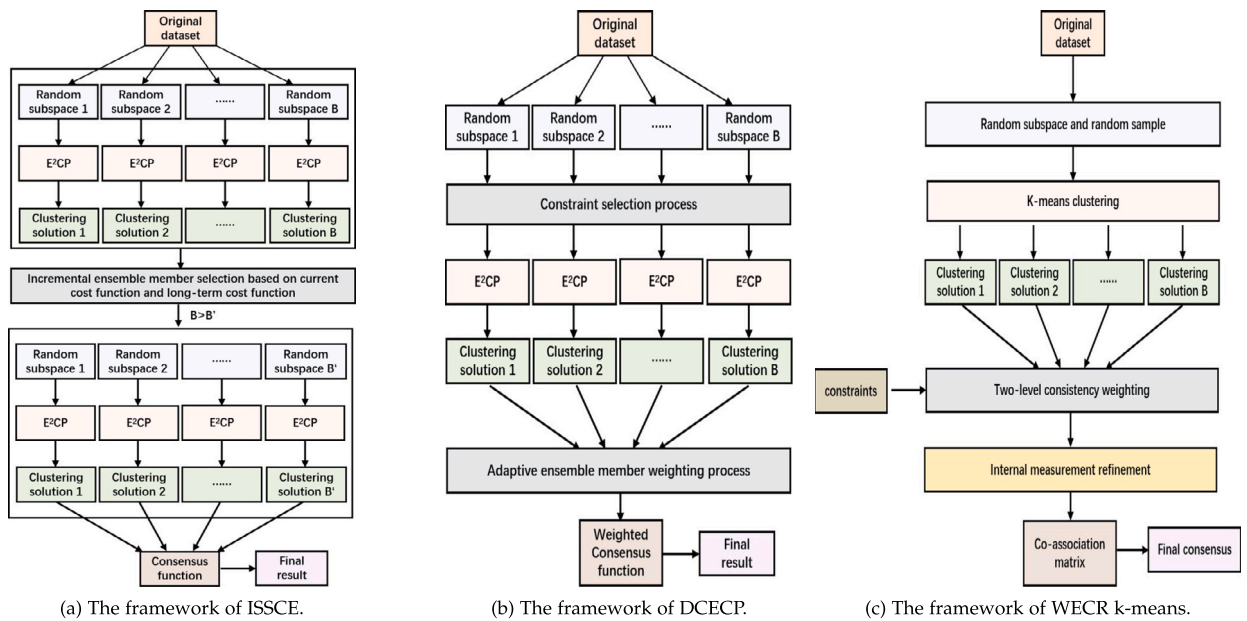
m: the dimension of the feature vectors.

p: the number of nearest neighbors on the graph.

B/B': the number of random subspace, selected subspace.

G/C: the number of maximum iterations of adaptive ensemble member weighting process/constraints.

s: the size of ensemble.

**Fig. 20.** Random subspace technique-based SSC.

by the random subspace technique and among the reduced space, the underlying architecture of the data set can be discovered. (2) *ISSCE* obtains a unified clustering solution by integrating multiple clustering solutions generated from different subspace, which provides more accurate, robust and stable final results. While the computation time is longer for *ISSCE* than single *SSC* approach, it generates final results of higher quality, which represents a worthwhile trade-off.

CESCP, [162] was another proposed *SSC* ensemble method using the random subspace technique, this method based on the technology of selected constraint projection. In addition to adopt the random subspace technique, *CESCP* uses the constraint

Table 7
Comparisons of evaluation measures.

Method	Range of values	Truth labels	Suitable for unbalanced datasets	Main idea
NMI(Normalized Mutual Information)	[0,1]	✓	✗	The similarity between two labels given to the same sample.
ACC(Accuracy)	[0,1]	✓	✗	The percentage of correct samples to the total number of samples.
F-measure	[0,1]	✓	✓	Balancing Precision and Recall, so that both can reach the highest index at the same time.
Precision	[0,1]	✓	✓	The percentage of correct samples to those with positive cluster labels.
Recall	[0,1]	✓	✓	The percentage of correct samples to those with positive true labels.
Purity	[0,1]	✓	✗	The number of correctly clustered samples divided by the total number of samples.
ARI(Adjusted Rand Index)	[-1,1]	✓	✓	It reflects the degree of overlap between the two divisions and ensures that the RI(Rand Index) value of the clustering result of the random division is close to 0.
MS(Minkowski Score)	[0,1]	✓	✗	The MS is the normalized distance between the two matrices. Lower MS implies better clustering solution.
XB(Xie–Beni index)	[0,1]	✗	✗	The overall average compactness and separation of a fuzzy c-partition.
Silhouette coefficient	[-1,1]	✗	✗	It measures the clustering results by determining the relative magnitude of cluster cohesion and cluster separation.
Pairwise F-measure	[0,1]	✓	✓	It evaluates clustering by considering same-cluster pairs and it is defined as the harmonic mean of pairwise precision and recall.
CRI(Constrained Rand Index)	[0,1]	✓	✓	It is a modification of the RI, and it confines its evaluation to decision which are determined by the constraints.

projection technique. Based on the technique of *CESCP*, *DCECP* is developed in [162]. *DCECP* is a double weighting semi-supervised ensemble clustering, including ensemble member weighting and constraint weighting. According to the difference of pairwise constraints subsets, each ensemble member is assigned to different weight values, so the ensemble's diversity is increased. As shown in Fig. 20(b), when compared with *RSSCE*, weighted constraint selection is performed in every random subspace generated in the first step. In contrast to *CESCP*, *DCECP* employs the process of *AEMW* to produce weight for ensemble members.

The experiments in [162] displays that *DCECP* is compared with various ensemble methods, such as *ISSCE*, *E²CPE*, [87], *RSKE*, [161], *HCCE*, [73]. The *NMI* value of *ISSCE* is 0.1 higher than *DCECP* on the dataset of *GLIOMA*. While on the remaining 14 datasets, *DCECP* achieves the best performance.

WECR k-means, [68] first utilized random sampling and random subspace techniques to convert the data from high-dimensional space to low dimensional space (as shown in Fig. 20(c)), then carried out conventional k-means on the obtained space to acquire the base partitions, finally, with the help of pairwise constraints, each cluster in the underlying partition was weighted to form the final result.

Based on the experiments in [68], *WECR k-means* is steady and performs the finest. The performance of *WECR k-means*, *E²CP* and *DCECP* degrades or fluctuations in varying degrees as the noise level gradually grows. As it reaches 20%, their performances are almost the same as unsupervised methods even on the normal datasets.

The random subspace technique-based *SSC* methods we surveyed are utilized into the ensemble framework. And the constraints are adding in the final clustering process. However, few methods consider prior information in the execution of the random subspace technique. The researches can focus on the random subspace technique itself in the future.

4. Evaluation measures

There are many evaluation measures to evaluate the performance of semi-supervised clustering, among which Normalized Mutual Information (*NMI*) [125], [157], Accuracy (*ACC*) [157], [132], Rand Index (*RI*) [133], Precision, Recall, F-measure [171] and Silhouette coefficient [119], etc. classical measures are frequently used for traditional unsupervised clustering. In addition to that, there are also some evaluation measures proposed for semi-supervised clustering, which will be described in detail in this section. In Table 7, we also explain the main idea of some commonly used evaluation indicators and analyze their merits and caveats.

For the evaluation measures proposed for semi-supervised clustering, Pairwise F-measure [9] and Constrained Rand Index (*CRI*) [66] are the most widely used. Pairwise F-measure [9] is defined by the harmonic mean of Pairwise Precision (eq. (26)) and Pairwise Recall (eq. (27)) as shown in eq. (28). Compared to the Precision, Recall and F-measure that evaluate the number of

correctly assigned samples, Pairwise Precision, Pairwise Recall and Pairwise F-measure evaluate the number of correctly assigned sample pairs.

$$\text{Pairwise Precision} = \frac{\# \text{Pairs Correctly Predicted In Same Cluster}}{\# \text{Total Pairs Predicted In Same Cluster}} \quad (26)$$

$$\text{Pairwise Recall} = \frac{\# \text{Pairs Correctly Actually In Same Cluster}}{\# \text{Total Pairs Predicted In Same Cluster}} \quad (27)$$

$$\text{Pairwise F-measure} = \frac{2 \times \text{Pairwise Precision} \times \text{Pairwise Recall}}{\text{Pairwise Precision} + \text{Pairwise Recall}} \quad (28)$$

Following RI, the modification Constrained Rand Index (CRI) [66] which is suitable for constrained clustering is defined. As the adding of constraints, the correctness of pairs fixed by the constraints or their closure is guaranteed.

$$\text{CRI} = \frac{\# \text{Correct free decisions}}{\# \text{Total free decisions}} \quad (29)$$

Both Pairwise F-measure and CRI need the number of real clusters in advance. Since the adding of constraints guarantees the correctness of pairs, CRI limits the evaluation to decisions that are less affected by constraints.

5. Applications of semi-supervised clustering

Semi-supervised clustering (SSC) is a hot topic and has been broadly adopted to medical, biological, business, journalism, financial fields and so on.

Medical field: With a large number of medical data storage in real-life databases, it is becoming more and more essential to build robust data mining procedures to analyze these data and retrieve the hidden knowledge behind it. As unsupervised clustering is not good at extracting accurate features, semi-supervised clustering is used by incorporating prior knowledge into traditional unsupervised clustering. The mainly applications of SSC in medical are diagnosis disease [155], [53], [167], [101], [70], early illness predict [21], [2], [34], [138], the analysis for injured patients [65] and analyzes biomedical documents [45], [13]. To solve the problem of inefficient data feature based on single-view, [167] proposed a multi-view SSC framework for the diagnosis of Parkinson's disease (PD) with the brain magnetic resonance imaging (MRIs). Image segmentation based supervised learning (such as support vector machine or neural networks) requires a tremendous quantity of training data (labeled voxels), which is expensive and time-consuming. To handle it, [101] proposed a SSC for magnetic resonance brain image segmentation. This method first labels an image slice's voxels clusters, after that uses the obtained label information to classify the remaining by Gaussian Mixture Model. To avoid the limitation (forced filtering the available labeled samples to preserve just that with a valid label) caused by supervised clustering, [21] proposed a DISSFCM model for predicting Bipolar Disorder (BD) chronic mental illness with samples gathered from the patient's interaction to the smartphone. [65] developed a model of multi-objective optimization (a penalty function of k-medians and a root-mean-square error of cross-validated) to group injured patients. These groups are a unique pattern based on the medical service apply during the first week after injury in a traffic accident. [45] considered three various kinds of information: the medical subject heading (MeSH) - semantic (MS), global-content (GC), and the local-content (LC) into SSC to boost the performance of clustering. At the same time, the SSC method they used also addressed the limitation of linear combination of three different types of information. Currently, the continuous stream of documents is processed in a batch-based manner, however, the batch-based methods often extract the events for each batch of documents collected in different hours. The existing technology is not enough to meet a large number of different medical data processing. For further research, it is worth of proposing new SSC technology that can adapt different types of medical informatics for a moment.

Biological field: In biological field, SSC can be used for biological recognition [159], [111], [23], [57]. [23] proposed a relaxational metric adaptation method to handle the sensitive problem of unsupervised clustering in selection of appropriate distance metric, this method was nonparametric. To address the various change in image, the evolution of dynamic parameter adaptation optimization models in image clustering is considered as a potential directions of research as suitable parameters is very important. In addition, in protein networks, community detection is helpful to find proteins with similar biological functions [148], [90], [88], [35]. SSC is also utilized as a valuable exploratory technique that enables biologists to recognize meaningful potential associations between genes. The performance of unsupervised clustering depends on the data distribution and its intrinsic property. Therefore, extending traditional unsupervised clustering to SSC to analyze gene expression [1], [58], [85], [95], [89], [94], predict gene function [64], [91], [26], sequence gene [139] is vital. The first work to solve the gene expression data clustering problem using SSC technique was reported by [1]. They proposed a multi-objective optimization method, in which initially FCM was applied to generate the labeled data. As common clustering methods cannot offer a comprehensive approach that look into the three categories of annotations (biological process, molecular function, and cellular component) and the traditional clustering methods use random initialization may cause inconsistent cluster generation. For detecting dynamic communities, [89] proposed two evolutionary nonnegative matrix factorization frameworks. Then, based on the gene expression data, they employed the dynamic networks associated with breast cancer progression to demonstrate the effectiveness of the methods. [64] presented a novel framework called CluFA, the framework utilized a fuzzy c-means clustering to find the optimal clusters. [139] proposed a semi-supervised NMF model utilizing the marked genes of cells as prior knowledge in the clustering analysis. However, as discovering essential clusters between genes for future biological investigation is expected in genomic data, the inability to give an accurate quantity of clusters is a serious defect. Therefore, determining the clusters quantity accurately in the gene data is a future research hotspot.

Business field: Detecting different community according to geographical location or interests plays a vital role in the business field. Individuals and their relationships can be represented by the networks. Community structure is considered as an essential property of networks. Networks are modeled as graphs in the area of network analysis, each entity/link is represented by a vertex/an edge. SSC can apply topology information network [148], [90], [11], heterogeneous information network (HIN) [88], [35] or attributed heterogeneous information networks (AHIN) [77], [59] with prior information to explore the community structure more accurately. [148] designed a unified semi-supervised community detection method, it alleviates the influence of topology information sparsity and noise. Among a HIN, the objects belong to different types (i.e., papers, authors, and conferences) and the relationships between the objects are also different, [88] introduced the relation-path notion to determine the closeness among two data of same type, it handled the challenges of multi-relations and diverse semantic meanings in heterogeneous networks. Attributed heterogeneous information networks is an HIN with various object attributes. SCHAIN-IRAM, [77] effectively aggregated attribute-based similarities and link-based similarities via a weighting scheme, the given supervision constraints were added to iterative mutual update process to solve the problem of weighting scheme optimization. Although the above applications are well implemented, on evolving networks, co-clustering of distinct kinds nodes on AHINs is still a problem to be solved. By considering the changes of edges and attributes and modeling the development of the related relationship, further efforts may be made in the future after splitting an AHINs into multiple snapshots of the HINs.

Journalism field: Semi-supervised clustering in the area of news analysis mainly focus on grouping news document [80], [100], [55]. Such as [80] clustered news documents into events from vast streams of breaking news and developed event stories by connecting related events in growing trees in an online manner, [100] categorized Urdu news datasets using the proposed Urdu clustering documents model, in which Urdu language has its own morphology, syntax and semantic. It solves the problem that unsupervised clustering is hindered on overlapping text documents. It is worth of proposing a new framework to adapt the continuous stream of documents, whenever a new document stream comes in, the new framework can cluster new documents into new events or assign them to existing events.

Financial field: Semi-supervised clustering can analysis the financial risk [48], financial fraud detection [135], [22] in financial area. To obtain satisfactory results in the absence of sufficient distinguishability of financial data, [48] explored the semi-supervised clustering method for the financial data prediction, in which accurate predictions were expected with few tagged data. This method first converted the input labeled data to the global prior information, and propagated it to learn the posterior probability, then a label diffusion model was constructed. On fraud detection, it is a great challenge to achieve a satisfied performance by only utilizing labeled data because of very little labeled data can be obtained. To solve this problem, [135] expanded the labeled data through user's social relations and proposed a *SemiGNN* method. This method can obtain a better performance as it utilizes the user attributes and their relations. While the sources of financial risk and the means of financial fraud are undergoing great changes, the use of better accurate or adaptive semi-supervised clustering models to predict different financial risks and financial fraud is desperately needed.

6. The challenges and development directions for semi-supervised clustering

On the basis of the analysis of the above sections, this section summaries several challenges and analyses the development directions for semi-supervised clustering (SSC).

6.1. Model design

Referring to the review of SSC, we here discuss several essential directions for SSC.

(1) **Fusing multiple prior information.** In recent years, as the era of big data approaches, most of the data sets have different types, semi-supervised clustering with only one type of prior information cannot meet the demand of complex data sets. Therefore, considering both label information and pairwise constraint or other types of prior information into the process of semi-supervised clustering gets more and more important.

(2) **Combining with other techniques.** Although the research on semi-supervised clustering has made great progress and achievements, many techniques can be tried to combine with semi-supervised clustering to improve the accuracy of semi-supervised clustering. Such as combining semi-supervised clustering and CDNN networks. The newly published semi-supervised CDNN depends on back-propagation CNN and Feed-Forward CNN, and most of them are established with an ensemble system. The experiments in [47] have demonstrated that a supervised CDNN acquires higher performance than conventional methods based on CDNN. Although CDNN architectures show better performance on complex data sets, no studies attempt to calculate cluster loss to improve the accuracy by combining semi-supervised clustering and CDNN networks. In addition, they face the challenges of higher time and memory complexity. Based on the above analysis, it requires some researchers to devote themselves to this field.

(3) **A United architecture for SSC.** On the basis of the aforementioned discussions, there are at least seven main architectures for SSC tasks. At the same time, we have also analyzed the methods according to the type of prior information, how to incorporate the prior information and the main problem solved by the method. Accordingly, we can find that each architecture or method has its own strengths and weaknesses. Notably, there are some explorations proposing a combined model that incorporates the merits of various types of methods. Consequently, to explore a unified SSC framework that would have advantages in terms of both theoretical and experimental results is worthwhile.

(4) **Objective functions.** The rapid development of semi-supervised clustering techniques is facilitated by the improvement of existing objective functions. For semi-supervised hierarchical clustering methods, they have been reported by few researches in the

past few years, while the emergence of global objective functions of hierarchical clustering [118] in 2015 creates conditions for the development of semi-supervised clustering technology. Therefore, it is more convenient for researchers to further incorporate prior information into semi-supervised clustering by modifying the objective functions.

(5) **High-dimensional data processing.** For high-dimensional data set, as the density information might become rare in the high-dimensional data space, it is hard for density-based SSC to process high-dimensional data set. For shifting the original dataset to a lower one, feature extraction, data reduction and kernel function and other pre-process techniques are adopted. However, the lower dimensional feature representation may not guarantee the density information. From the perspective of manifold learning, some literature shows that many techniques can offer an acceptable solution. It requires further research by researchers.

(6) **Reducing computational complexity.** Many SSC methods need to compute a distance matrix for all pairs of data set, it causes a higher computational complexity. To simplify the calculation, a matrix factorization technique must be advanced further. In addition, resolving a large data set into several subsets is another method to be further researched.

(7) **Evaluation measure.** The evaluation criteria are very important for semi-supervised clustering. To judge whether the algorithm is good or not, the better evaluation criteria for semi-supervised clustering methods needs to be proposed.

6.2. Supervising information collection

Our discussion and analysis on current types of supervisory information revealed several factors that are critical to the future of supervisory information collection.

(1) **Constraints number.** The number of constraints is very limited because of its difficulty to collect. The researchers have shown that active learning is efficient [9], [169], [42]. However, these methods only use one type of constraint utility measures in the constraint selection procedure. In order to pick the maximally informative constraint, it is necessary to provide different types of constraint utility measures in the future work.

(2) **Space-level constraints.** The background information used recently in semi-supervised clustering is instance-level. However, they are inappropriate for semi-supervised hierarchical clustering as the data objects are linked at distinct levels. It is reasonable for researchers to dedicate in extending instance-level constraints to space-level constraints to get better clustering performance.

6.3. Efficient SSC for real-world application

Based on the above elaboration on the application of semi-supervised clustering, we summarize the challenges and development prospects of the application of semi-supervised clustering.

The current leading SSC models are designed to be complex in order to achieve increased learning capabilities and improved performance. However, to meet the requirements of applications, more clever and lightweight architectures are needed. In addition, the existing techniques are not sufficient for the large amount of different types of data generated in various fields. In some specific fields, there is an urgent need to use more accurate or adaptive semi-supervised clustering models. Such as in the financial field, more accurate or adaptive semi-supervised clustering models is needed as the sources of financial risk and the means of financial fraud are changing dramatically. For further research, it is worthwhile to propose new SSC techniques.

7. Conclusions

In recent decades, the semi-supervised clustering technique (SSC) has been rapidly evolving, improving the performance of clustering as it solves the problem that the results obtained from unsupervised clustering may be highly inconsistent with the true partition. In many fields, semi-supervised clustering is also of practical importance. Although semi-supervised clustering is getting a growing attention by researchers and the literature has been greatly enriched, it remains difficult to link them. To choose a suitable method on the issue in hand and determine the work that remains to be completed in the topic, a comprehensive survey of SSC is presented in this paper.

First, the existing methods are grouped into seven main categories in our work, including partition-based SSC, hierarchical-based SSC, density-based SSC, graph-based SSC, neural network-based SSC, Nonnegative Matrix Factorization-based SSC and random subspace technique-based SSC. And these seven kinds of SSC methods and their subcategories are reviewed and discussed in detail. Based on the existing methods, the applications in different fields are summarized and analyzed. At last, the challenges and promising directions for future efforts for SSC is revealed. For the beginners, this survey can offer a thorough understanding of SSC methods. For the researchers, through a clear description of the research progress and some challenges faced nowadays in this subject, they can conduct further studies to propose more adaptive and advanced SSC methods to address the unsolved problems in SSC, so as to promote the development of the field related to SSC.

In conclusion, SSC has made notable progress. However, several issues remain under-explored in terms of achieving more effective clustering performance. For partition-based SSC and density-based SSC, most of them are not suitable to high-dimensional dataset, while the semi-supervised fuzzy c-means in partition-based SSC and density-based SSC can recognize clusters of arbitrary shapes. Nonnegative matrix factorization-based SSC can deal with the dataset with high-dimensional. However, the majority of the methods take high time complexity, so does the graph-based SSC. Considering the specificity of the clustering process of hierarchical-based SSC, space-level constraints and cluster-level constraints can achieve better performance. We anticipate that this survey will offer a valid way to comprehend the state-of-the-arts at present and provide an insight for further exploring the SSC.

CRediT authorship contribution statement

Jianghui Cai: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Software, Validation, Writing – original draft. **Jing Hao:** Conceptualization, Formal analysis, Investigation, Methodology, Software, Visualization, Writing – original draft. **Haifeng Yang:** Conceptualization, Funding acquisition, Project administration, Resources, Supervision, Validation, Writing – original draft, Writing – review & editing. **Xujun Zhao:** Data curation, Supervision. **Yuqing Yang:** Resources, Software, Visualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Appendix A

Abbreviation	Full name
SLINK/CLINK	single link/complete link
OPTICS	ordering points to identify the clustering structure
DNMF	discriminative nonnegative matrix factorization
DCECP	double weighting semi-supervised ensemble clustering based on selected constraint projection
Cop-Kmeans	constrained K-means
ICop-Kmeans	improved Cop-Kmeans
CVQE	constrained vector quantization error
LCVQE	the modification of CVQE
PCKmeans	pairwise constraints k-means
MK-MEANS	K-Means clustering with the metric learning component
MPCKM	metric pairwise constrained k-means
ASSKMM	an adaptive kernel method for semi-supervised clustering
SCKMM	semi-supervised clustering kernel method based on metric learning
PCSKM	pairwise constrained sparse k-means
CSSC-Kmeans	compact-cluster-based SSC K-means
FCM	fuzzy c-means
SSKFCM	semi-supervised kernel fuzzy c-mean algorithm
FSREC-PC	a novel fuzzy symmetric relative entropy clustering with pairwise-constraints
SMKFC-ER	semi-supervised multiple kernel fuzzy clustering based on entropy and relative entropy
SFC-ER	semi-supervised fuzzy clustering based on entropy and relative entropy called
SMUC	semi-supervised metric-based fuzzy clustering
AFCC	active fuzzy constrained clustering
sSFCM	semi-supervised standard fuzzy c-means clustering
eSFCM	semi-supervised entropy regularized fuzzy c-means clustering
SSFCM-ID	semi-supervised fuzzy c-means clustering algorithm for incomplete data

(continued on next page)

Abbreviation	Full name
LHC-S3FCM	local homogeneous consistent safe semi-supervised clustering
CS3FCM	confidence-weighted safe semi-supervised
SSFCM-ID	a semi-supervised fuzzy c-means clustering algorithm for incomplete data
CrKMd	credibilistic k-mode
CrKMd-KNN	CrKMd integrates with K nearest neighbor
CrKMd-ANN	CrKMd integrates with artificial neural network
CrKMd-DT	CrKMd integrates with decision tree
CrKMd-RF	CrKMd integrates with random forest
MLCrKMd	machine learning integrated semi-supervised clustering
IPoptim/UltraTran	the iterative projection algorithm/the transitive dissimilarity transformation
HAC	standard agglomerative hierarchical clustering
HACoc	constraint-based standard agglomerative hierarchical clustering
HCAC	hierarchical confidence-Based active clustering
CCL	constrained complete-link
CECH	clustering ensemble based on CHAMELEON
SCECH	semi-supervised clustering ensemble based on CHAMELEON
SCEV	semi-supervised clustering ensemble by voting
SS-CSPA	semi-supervised cluster-based similarity partitioning algorithm
SSCC	semi-supervised consensus clustering
DBSCAN	density based spatial clustering of Applications with Noise
HISSCLU	a hierarchical density-based method for semi-supervised clustering
SSDBSCAN	semi-supervised DBSCAN
SSDC	semi-supervised DenPeak clustering
C-DBSCAN	constraint-based variation of DBSCAN
NSDR	near stranger or distant relative
TI-APJCF and TII-APJCF	type-I and type-II affinity and penalty jointly constrained spectral clustering
NLPPC	new label propagation with pairwise constraints
HMRF	hidden markov random fields
PNLP	the positive and negative label propagation algorithm
PCLP	label propagation with pairwise constraints
SC-PC	spectral clustering method based on pairwise constraints
SSGC	a new semi-supervised graph based clustering
AMSS	an auto-weighted multi-view framework for semi-supervised
GAN	Generative Adversarial Nets
MGSC	semi-supervised learning for multiple graphs by gradient flow method
RSSC	robust semi-supervised spectral clustering
SSCCDS	semi-supervised sparse neighbor constrained co-clustering with dissimilarity and similarity regularization model
CSSC-SC	compact-cluster-based SSC spectral clustering
AMGL	auto-weighted multiple graph learning
ss- S^{2c^2}	semi-supervised siamese classifiers for clustering
DEC	deep embedded clustering
SDEC	semi-supervised deep embedded clustering
IDEC	improved deep embedded clustering
NMF	nonnegative matrix factorization
CNMF	constrained nonnegative matrix factorization
DNMF	discriminative nonnegative matrix factorization

Abbreviation	Full name
GDNMF	semi-supervised graph-based discriminative nonnegative matrix factorization
GNMF	graph-based nonnegative matrix factorization
PGCNMF	pairwise constrained graph regularized convex nonnegative matrix factorization
CSNMF	correntropy based semi-supervised nonnegative matrix factorization
CPSNMF	semi-supervised NMF via constraint propagation
CMVMNF	constrained multi-view nonnegative matrix factorization
MVCNMF	constrained nonnegative matrix factorization with sparseness constraint
MVOCNMF	semi-supervised multi-view clustering based on orthonormality-constrained NMF
DHPS-NMF	dual hypergraph regularized partially shared non-negative matrix factorization
semiEMC	a semi-supervised exclusive multiview clustering
RSSCE	random subspace based semi-supervised clustering ensemble framework
E^2 CP	the constraint propagation approach
AEMW	adaptive ensemble member weighting
ISSCE	incremental semi-supervised clustering ensemble framework
CESCP	semi-supervised clustering ensemble based on selected constraint projection
DCECP	double weighting semi-supervised ensemble clustering based on selected constraint projection
E^2 CPE	exhaustive and efficient constraint propagation clustering ensemble
RSKE	random K-means clustering ensemble
HCCE	hierarchical clustering ensemble
WECR k-means	weighted consensus of random k-means ensemble
DISSFCM	dynamic incremental semi-supervised fuzzy c-means
CluFA	clustering functional annotation
CNN	convolutional neural network
CDNN	convolutional deep neural network

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